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# SECR1013 DIGITAL LOGIC

## MODULE 8b: COUNTERS (SYNC)

FACULTY OF COMPUTING



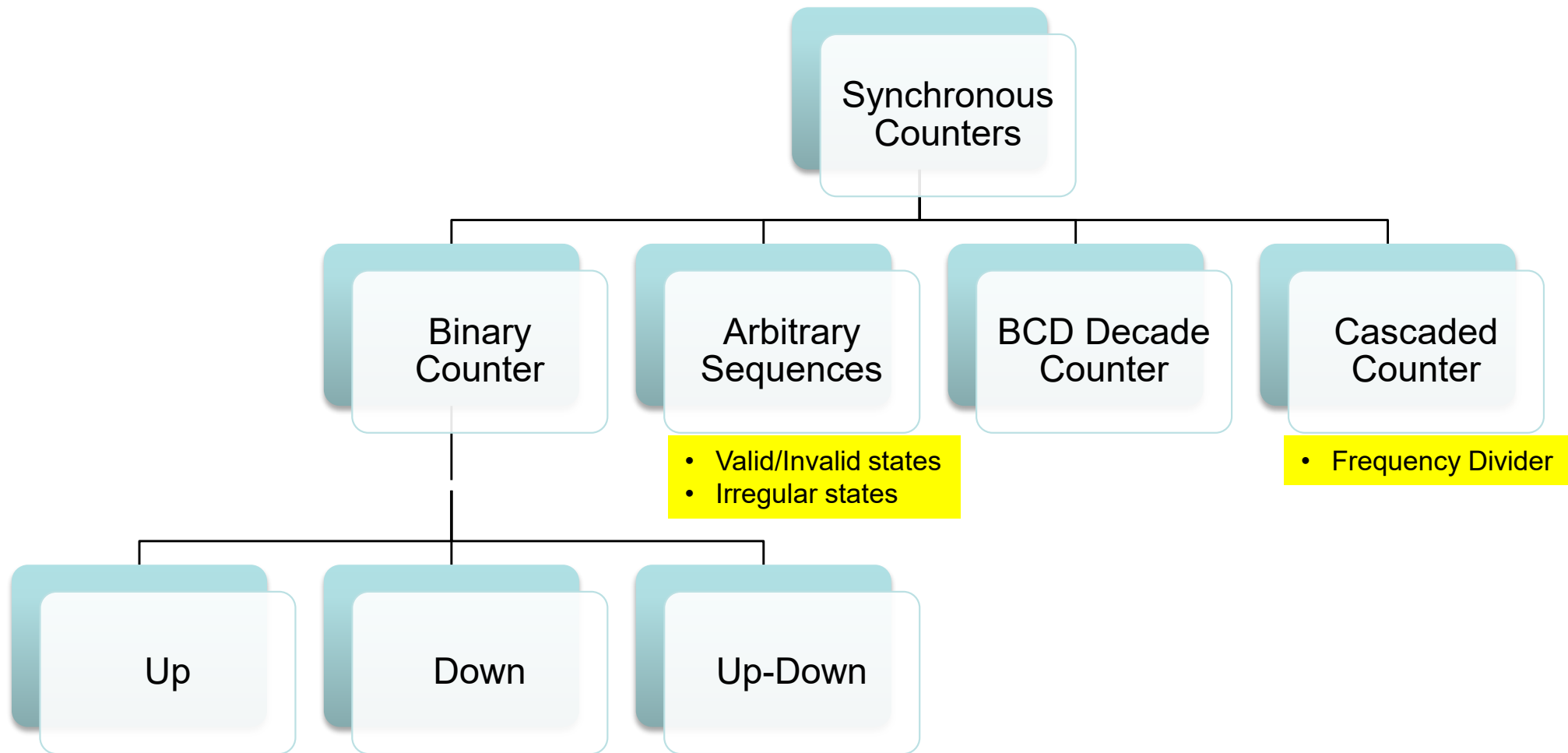
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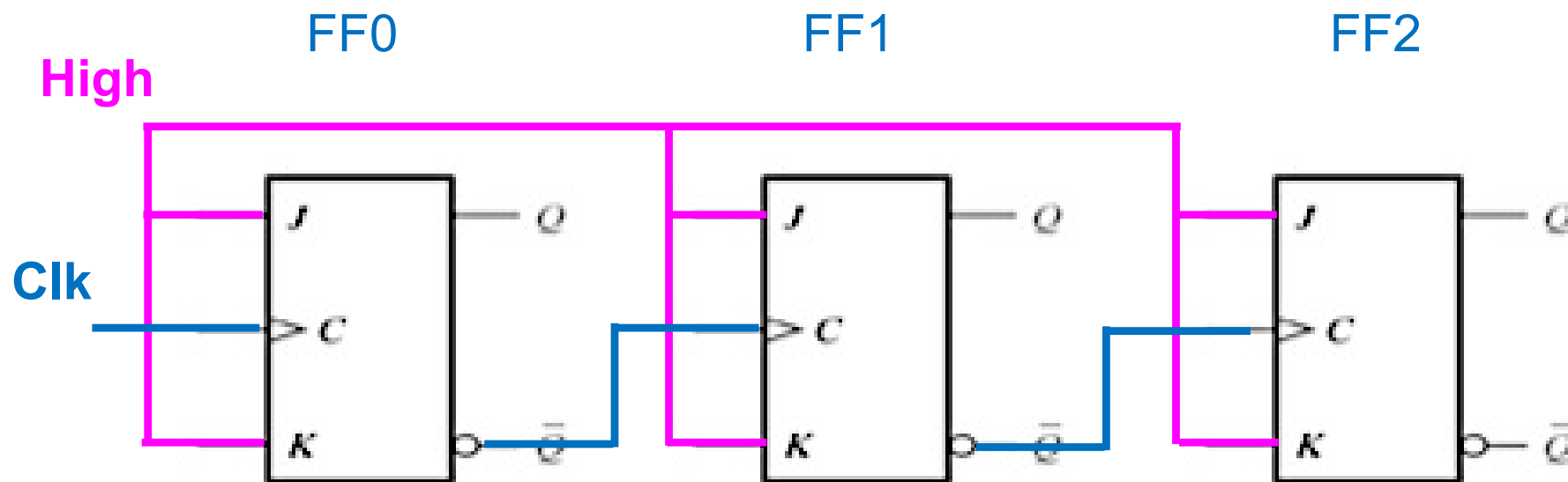
# Synchronous Counters:

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RECAP

## Design 3-bit Count Up Asynchronous Counter using J-K Flip-flop

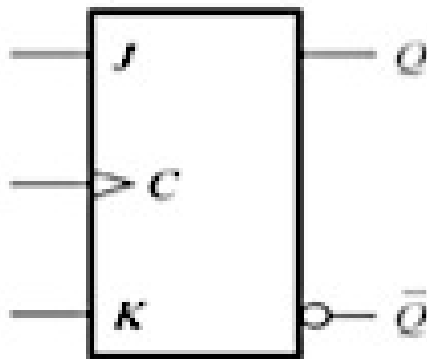


Async counter is  
easy, straight  
forward to design

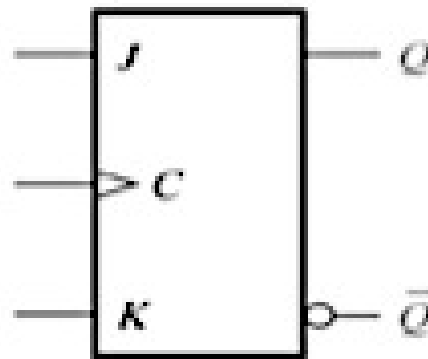
How ...

# Design 3-bit Count Up Synchronous Counter using J-K Flip-flop

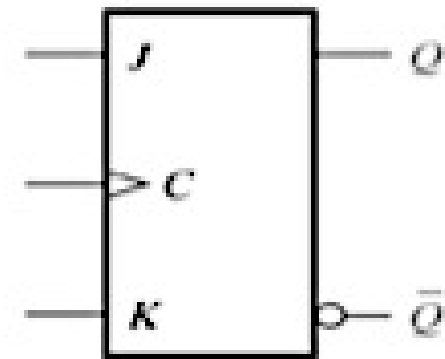
FF0



FF1



FF2



Sync counter is **not**  
straight forward to  
design!

# Synchronous Counter Design

- **Step 1**  
Describe a general sequential circuit in terms of its basic parts and its input and outputs.
- **Step 2**  
Develop state diagram.
- **Step 3**  
Create next state table.
- **Step 4**  
Create flip-flop transition table.
- **Step 5**  
Use K-maps to derive the logic equations.
- **Step 6**  
Implement counter implementation.



Excitation Table

Truth table for a positive edge-triggered J-K flip-flop.

| Inputs |     |            | Outputs     |             | Comments  |
|--------|-----|------------|-------------|-------------|-----------|
| $J$    | $K$ | CLK        | $Q$         | $\bar{Q}$   |           |
| 0      | 0   | $\uparrow$ | $Q_0$       | $\bar{Q}_0$ | No change |
| 0      | 1   | $\uparrow$ | 0           | 1           | RESET     |
| 1      | 0   | $\uparrow$ | 1           | 0           | SET       |
| 1      | 1   | $\uparrow$ | $\bar{Q}_0$ | $Q_0$       | Toggle    |

- JK Flip-Flop

Refer to JK FF truth table

| FF State |   | Present State | Next State |
|----------|---|---------------|------------|
| J        | K | $Q_n$         | $Q_{n+1}$  |
| 0        | 0 | 0             | 0          |
| 0        | 0 | 1             | 1          |
| 0        | 1 | 0             | 0          |
| 0        | 1 | 1             | 0          |
| 1        | 0 | 0             | 1          |
| 1        | 0 | 1             | 1          |
| 1        | 1 | 0             | 1          |
| 1        | 1 | 1             | 0          |

NO CHANGE

RESET

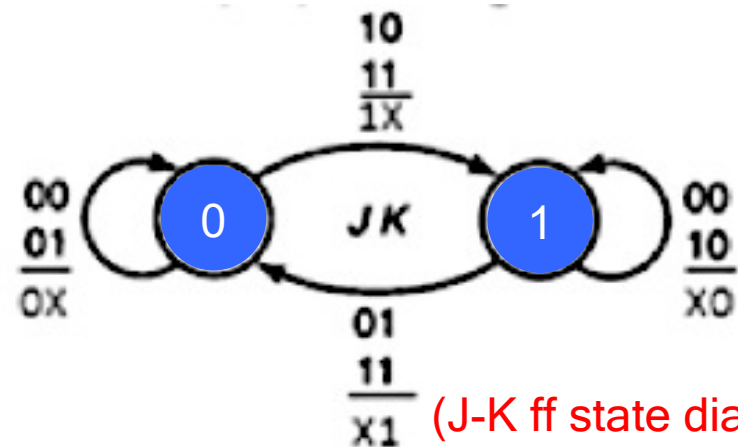
SET

TOGGLE

# FF Excitation Table

About creating Transition table

$J, K = 0/1 = X$   
\*don't care



(J-K ff state diagram)

Truth table for a positive edge-triggered J-K flip-flop.

| Inputs |     |            | Outputs     |             | Comments  |
|--------|-----|------------|-------------|-------------|-----------|
| $J$    | $K$ | CLK        | $Q$         | $\bar{Q}$   |           |
| 0      | 0   | $\uparrow$ | $Q_0$       | $\bar{Q}_0$ | No change |
| 0      | 1   | $\uparrow$ | 0           | 1           | RESET     |
| 1      | 0   | $\uparrow$ | 1           | 0           | SET       |
| 1      | 1   | $\uparrow$ | $\bar{Q}_0$ | $Q_0$       | Toggle    |

- JK Flip-Flop

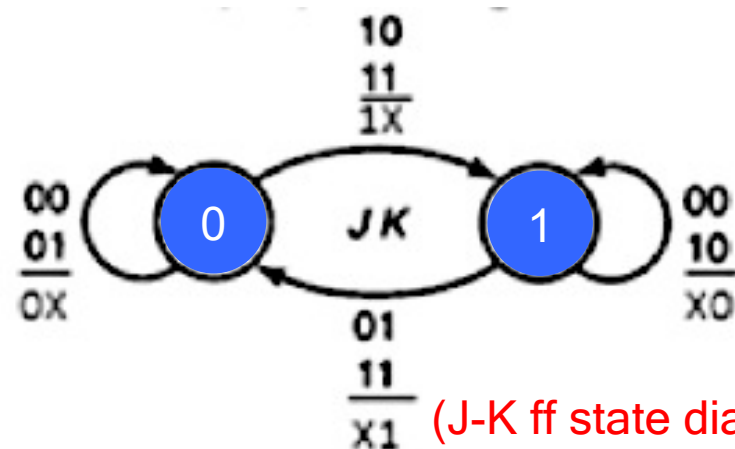
Refer to JK FF truth table

| FF State  |   | Present State | Next State |
|-----------|---|---------------|------------|
| J         | K | $Q_n$         | $Q_{n+1}$  |
| NO CHANGE | 0 | 0             | 0          |
|           | 0 | 1             | 1          |
| RESET     | 1 | 0             | 0          |
|           | 1 | 1             | 0          |
| SET       | 0 | 0             | 1          |
|           | 0 | 1             | 1          |
| TOGGLE    | 1 | 0             | 1          |
|           | 1 | 1             | 0          |

## FF Excitation Table

### JK-ff Excitation Table

| Present State | Next State | FF State |   |
|---------------|------------|----------|---|
| $Q_n$         | $Q_{n+1}$  | J        | K |
| 0             | 0          | 0        | X |
| 0             | 1          | 1        | X |
| 1             | 0          | X        | 1 |
| 1             | 1          | X        | 0 |



(J-K ff state diagram)

### Exercise 8b.1:

Construct the  
excitation table  
for **D flip-flop**

(using its state  
diagram)

### Solution:

D flip-flop excitation table

| Present<br>State | Next<br>State | FF<br>State |
|------------------|---------------|-------------|
| $Q_n$            | $Q_{n+1}$     | D           |
|                  |               |             |
|                  |               |             |
|                  |               |             |
|                  |               |             |

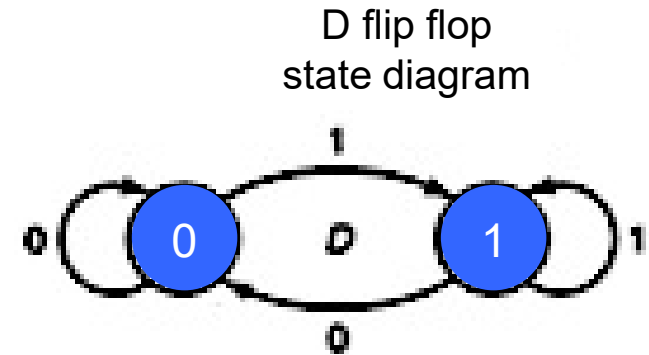
## Exercise 8b.1:

Construct the  
excitation table  
for **D flip-flop**

(using its state  
diagram)

| D | Present state<br>$Q_n$ | Next state<br>$Q_{n+1}$ |
|---|------------------------|-------------------------|
| 0 | 0                      | 0                       |
| 0 | 1                      | 0                       |
| 1 | 0                      | 1                       |
| 1 | 1                      | 1                       |

*Truth table of D flip flop*



D flip-flop excitation table

| Present State | Next State | FF State |
|---------------|------------|----------|
| $Q_n$         | $Q_{n+1}$  | D        |
| 0             | 0          | 0        |
| 0             | 1          | 1        |
| 1             | 0          | 0        |
| 1             | 1          | 1        |

≡

| $Q_{n+1}$ | D |
|-----------|---|
| 0         | 0 |
| 1         | 1 |

### Exercise 8b.2:

Construct the  
excitation table  
for **T flip-flop**

(using its state  
diagram)

### Solution:

T flip-flop excitation table

| Present<br>State | Next<br>State | FF<br>State |
|------------------|---------------|-------------|
| $Q_n$            | $Q_{n+1}$     | T           |
|                  |               |             |
|                  |               |             |
|                  |               |             |
|                  |               |             |

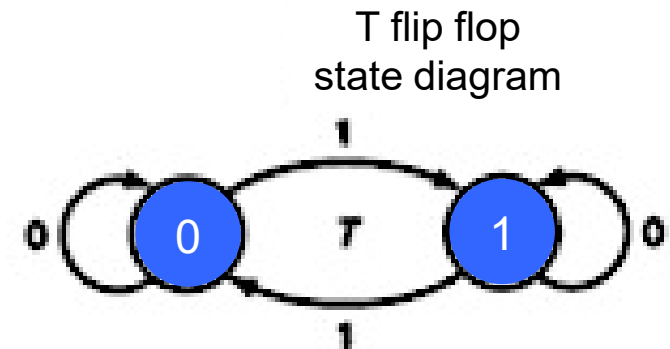
## Exercise 8b.2:

Construct the  
excitation table  
for **T flip-flop**

(using its state  
diagram)

| T | Q | Q <sub>+1</sub> | State     |
|---|---|-----------------|-----------|
| 0 | 0 | 0               | NO CHANGE |
| 0 | 1 | 1               |           |
| 1 | 0 | 1               | TOGGLES   |
| 1 | 1 | 0               |           |

*Truth Table of T flip flop*



T flip-flop excitation table

| Present State | Next State | FF State |
|---------------|------------|----------|
| $Q_n$         | $Q_{n+1}$  | T        |
| 0             | 0          | 0        |
| 0             | 1          | 1        |
| 1             | 0          | 1        |
| 1             | 1          | 0        |

≡

| $Q_{n+1}$        | T |
|------------------|---|
| $Q_n$            | 0 |
| $\overline{Q_n}$ | 1 |

### Note:

These **excitation tables** will be used while filling in the flip-flop **transition table** in **STEP 4** of designing synchronous counter.

## Summary

Excitation tables of flip-flops:

NO CHANGE / RESET

SET / TOGGLE

RESET / TOGGLE

NO CHANGE / SET

| Present State | Next State | FF State |   |
|---------------|------------|----------|---|
| $Q_n$         | $Q_{n+1}$  | J        | K |
| 0             | 0          | 0        | X |
| 0             | 1          | 1        | X |
| 1             | 0          | X        | 1 |
| 1             | 1          | X        | 0 |

J-K flip-flop

| $Q_{n+1}$ | D |
|-----------|---|
| 0         | 0 |
| 1         | 1 |

D flip-flop

| $Q_{n+1}$        | T |
|------------------|---|
| $Q_n$            | 0 |
| $\overline{Q_n}$ | 1 |

T flip-flop



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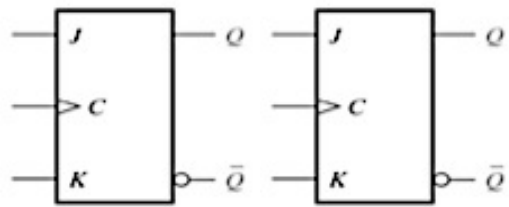
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# Synchronous Counters:

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Up

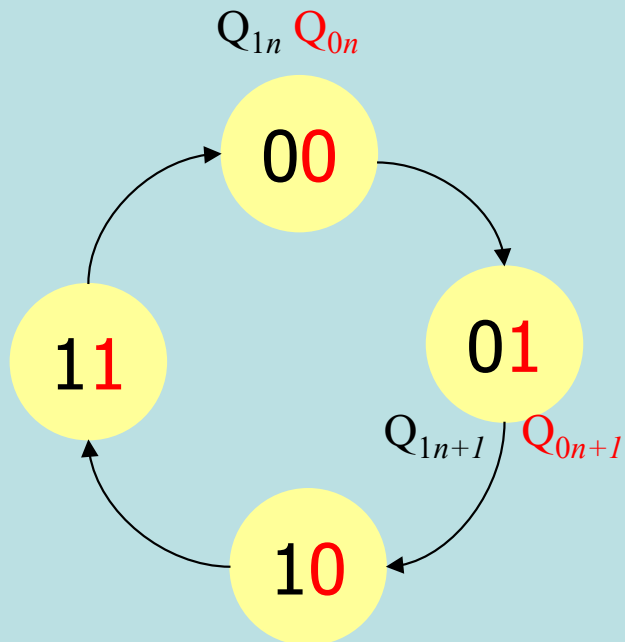


# Design 2-bit Up Synchronous Counter:

## J-K Flip-flop

- **Step 1:**  
To design 2-bit up synchronous counter using JK FF.

- **Step 2:** Draw state diagram



- **Step 3:** Create next state table

| Present State |          | Next State |            |
|---------------|----------|------------|------------|
| $Q_{1n}$      | $Q_{0n}$ | $Q_{1n+1}$ | $Q_{0n+1}$ |
| 0             | 0        | 0          | 1          |
| 0             | 1        | 1          | 0          |
| 1             | 0        | 1          | 1          |
| 1             | 1        | 0          | 0          |

**Note:**

While filling in the flip-flop transition table, refer to the excitation table.

| Present State | Next State | FF State |   |
|---------------|------------|----------|---|
| $Q_n$         | $Q_{n+1}$  | J        | K |
| 0             | 0          | 0        | X |
| 0             | 1          | 1        | X |
| 1             | 0          | X        | 1 |
| 1             | 1          | X        | 0 |

- Step 4:

Construct the flip-flop transition table

FF1

FF0

| Present State |          | Next State |            | JK Transition |       |       |       |
|---------------|----------|------------|------------|---------------|-------|-------|-------|
| $Q_{1n}$      | $Q_{0n}$ | $Q_{1n+1}$ | $Q_{0n+1}$ | $J_1$         | $K_1$ | $J_0$ | $K_0$ |
| 0             | 0        | 0          | 1          |               |       |       |       |
| 0             | 1        | 1          | 0          |               |       |       |       |
| 1             | 0        | 1          | 1          |               |       |       |       |
| 1             | 1        | 0          | 0          |               |       |       |       |

**Note:**

While filling in the flip-flop transition table, refer to the excitation table.

| Present State | Next State | FF State |   |
|---------------|------------|----------|---|
| $Q_n$         | $Q_{n+1}$  | J        | K |
| 0             | 0          | 0        | X |
| 0             | 1          | 1        | X |
| 1             | 0          | X        | 1 |
| 1             | 1          | X        | 0 |

- Step 4:

Construct the flip-flop transition table

FF1

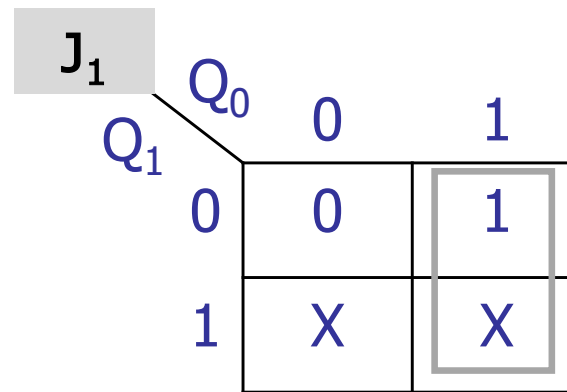
FF0

| Present State |          | Next State |            | JK Transition |       |       |       |
|---------------|----------|------------|------------|---------------|-------|-------|-------|
| $Q_{1n}$      | $Q_{0n}$ | $Q_{1n+1}$ | $Q_{0n+1}$ | $J_1$         | $K_1$ | $J_0$ | $K_0$ |
| 0             | 0        | 0          | 1          | 0             | X     | 1     | X     |
| 0             | 1        | 1          | 0          | 1             | X     | X     | 1     |
| 1             | 0        | 1          | 1          | X             | 0     | 1     | X     |
| 1             | 1        | 0          | 0          | X             | 1     | X     | 1     |

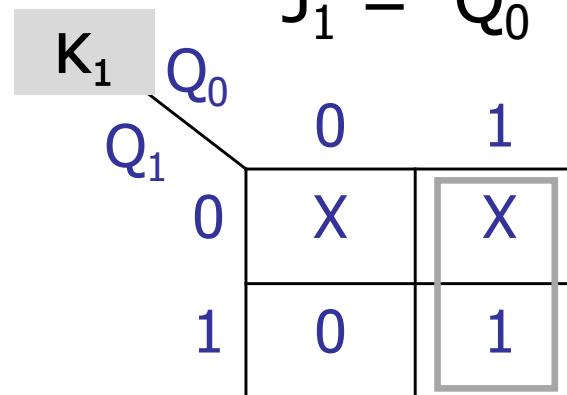
| Present State |          | Next State |            | JK Transition |       |       |       |
|---------------|----------|------------|------------|---------------|-------|-------|-------|
| $Q_{1n}$      | $Q_{0n}$ | $Q_{1n+1}$ | $Q_{0n+1}$ | $J_1$         | $K_1$ | $J_0$ | $K_0$ |
| 0             | 0        |            |            | 0             | X     | 1     | X     |
| 0             | 1        |            |            | 1             | X     | X     | 1     |
| 1             | 0        |            |            | X             | 0     | 1     | X     |
| 1             | 1        |            |            | X             | 1     | X     | 1     |

- Step 5:  
Use K-maps to  
derive the logic  
equations

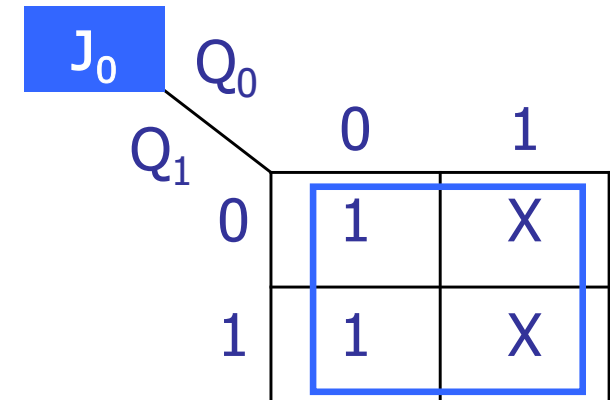
(for present state  
only).



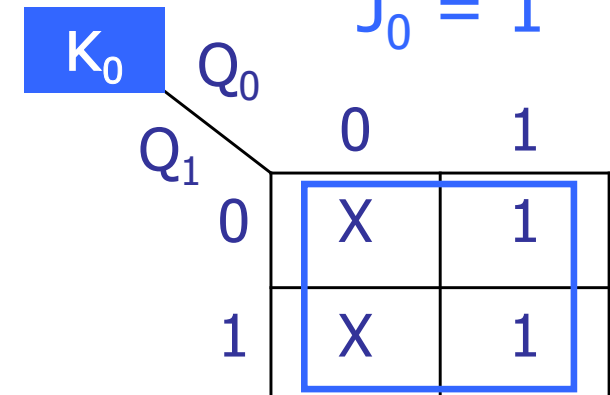
$$J_1 = Q_0$$



$$K_1 = Q_0$$

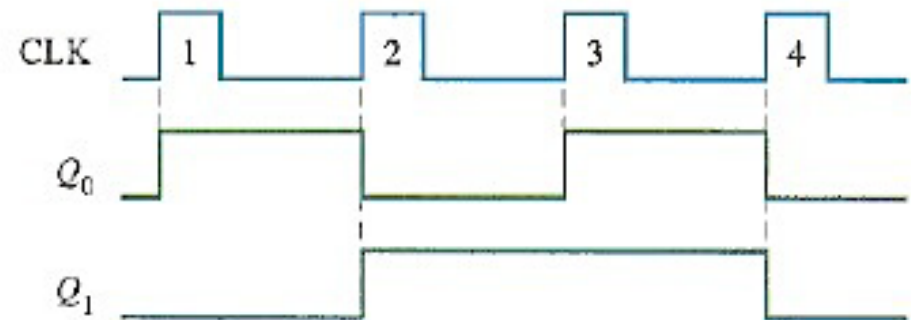
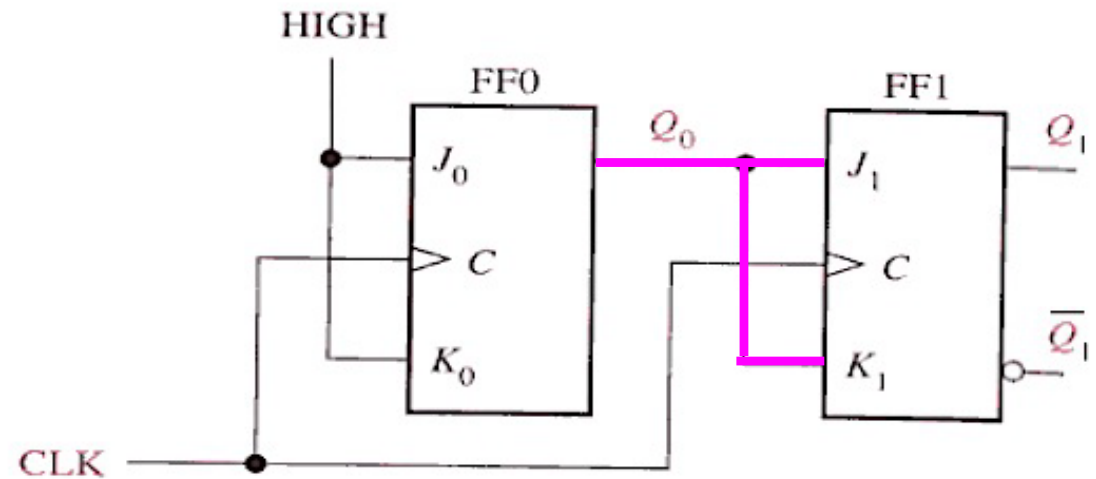
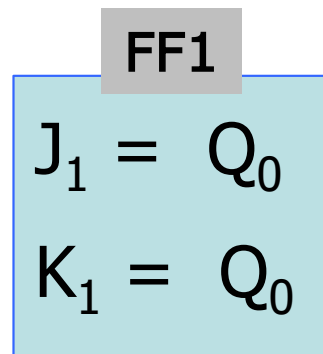
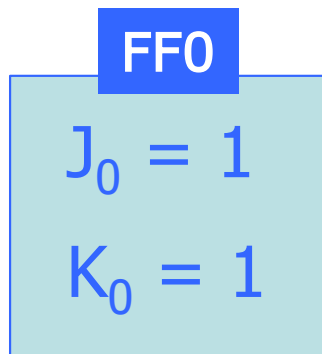


$$J_0 = 1$$



$$K_0 = 1$$

- **Step 6:** Implement counter by drawing the logic symbol connection / counter circuit.



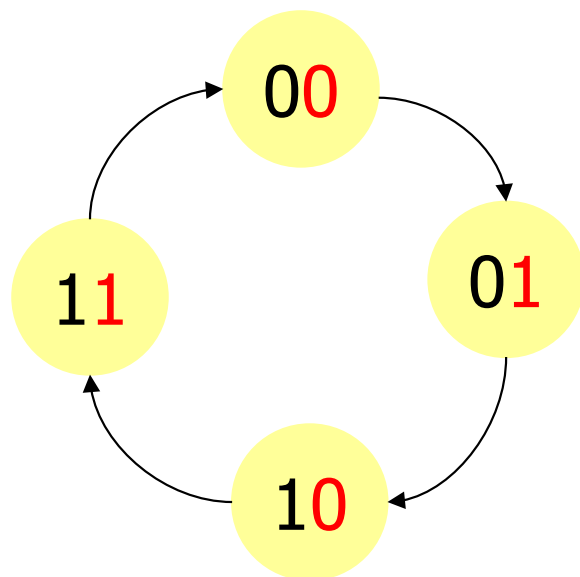
$Q_1$  toggle when  $Q_0=1$

- $Q_0$  toggle at positive edge (CLK1, CLK2, CLK3, CLK4)
- $Q_1$  toggle when  $Q_0=1$  at positive edge (CLK2, CLK4)

**Exercise 8b.3:** Design 2-bit up synchronous counter that using D flip-flop with positive edge triggered. Show all steps clearly.

## Solution 8b.3:

- Step 2: Draw state diagram



- Step 3: Create next state table

| Present State |       | Next State |          |
|---------------|-------|------------|----------|
| $Q_1$         | $Q_0$ | $Q_{1+}$   | $Q_{0+}$ |
| 0             | 0     | 0          | 1        |
| 0             | 1     | 1          | 0        |
| 1             | 0     | 1          | 1        |
| 1             | 1     | 0          | 0        |

**Note:**

While filling in the flip-flop transition table, refer to the excitation table.

| $Q_{n+1}$ | D |
|-----------|---|
| 0         | 0 |
| 1         | 1 |

- **Step 4:**

Construct flip-flop transition table using D excitation table.

| Present State |       | Next State |          | D Transition |       |
|---------------|-------|------------|----------|--------------|-------|
| $Q_1$         | $Q_0$ | $Q_{1+}$   | $Q_{0+}$ | $D_1$        | $D_0$ |
| 0             | 0     | 0          | 1        | 0            | 1     |
| 0             | 1     | 1          | 0        | 1            | 0     |
| 1             | 0     | 1          | 1        | 1            | 1     |
| 1             | 1     | 0          | 0        | 0            | 0     |

**Note:**

While filling in the flip-flop transition table, refer to the excitation table.

| $Q_{n+1}$ | D |
|-----------|---|
| 0         | 0 |
| 1         | 1 |

- Step 4:

Construct flip-flop transition table using the excitation table.

*(for present state only).*

| Present State |       | Next State |          | D Transition |       |
|---------------|-------|------------|----------|--------------|-------|
| $Q_1$         | $Q_0$ | $Q_{1+}$   | $Q_{0+}$ | $D_1$        | $D_0$ |
| 0             | 0     |            |          | 0            | 1     |
| 0             | 1     |            |          | 1            | 0     |
| 1             | 0     |            |          | 1            | 1     |
| 1             | 1     |            |          | 0            | 0     |

- Step 5:

Use K-maps to derive the logic equations.

**D<sub>0</sub>**

|       |       |   |   |
|-------|-------|---|---|
| $Q_1$ | $Q_0$ | 0 | 1 |
|       | 0     | 1 | 0 |
| 1     | 1     | 1 | 0 |

$$D_0 = \bar{Q}_0$$

**D<sub>1</sub>**

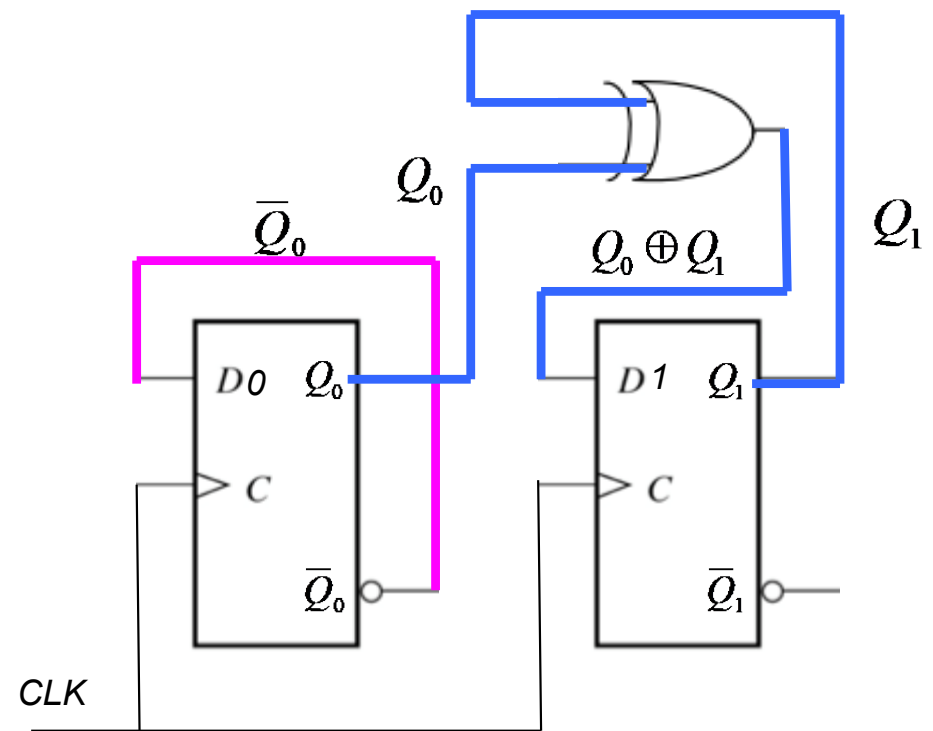
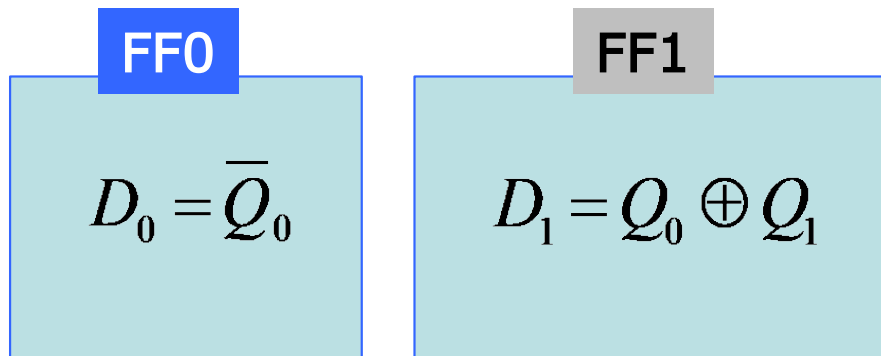
|       |       |   |   |
|-------|-------|---|---|
| $Q_1$ | $Q_0$ | 0 | 1 |
|       | 0     | 0 | 1 |
| 1     | 1     | 1 | 0 |

$$D_1 = Q_0 \bar{Q}_1 + \bar{Q}_0 Q_1$$

$$= Q_0 \oplus Q_1$$

(XOR Page: 97, 152)

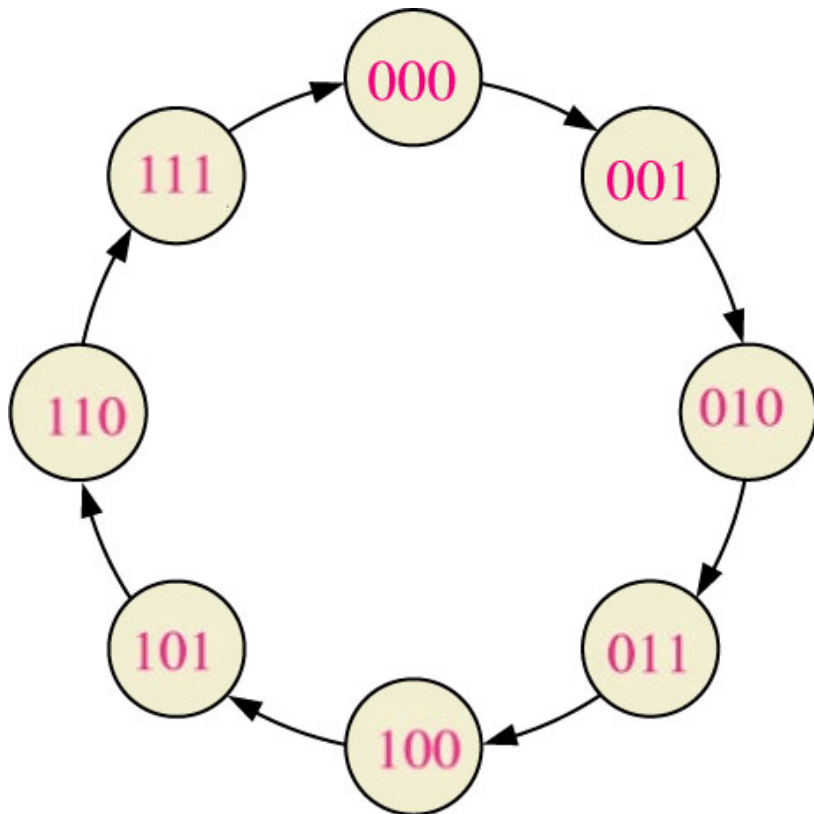
- **Step 6:** Implement counter implementation by drawing the logic symbol connection / counter circuit.



**Exercise 8b.4:** Design 3-bit up synchronous counter that using T flip-flop with positive edge triggered. Show all steps clearly.

# Design 3-bit Up Synchronous Counter: J-K Flip-flop

- Step 2: State diagram



- Step 3: Next State table

| Present State |       |       | Next State |          |          |
|---------------|-------|-------|------------|----------|----------|
| $Q_2$         | $Q_1$ | $Q_0$ | $Q_{2+}$   | $Q_{1+}$ | $Q_{0+}$ |
| 0             | 0     | 0     | 0          | 0        | 1        |
| 0             | 0     | 1     | 0          | 1        | 0        |
| 0             | 1     | 0     | 0          | 1        | 1        |
| 0             | 1     | 1     | 1          | 0        | 0        |
| 1             | 0     | 0     | 1          | 0        | 1        |
| 1             | 0     | 1     | 1          | 1        | 0        |
| 1             | 1     | 0     | 1          | 1        | 1        |
| 1             | 1     | 1     | 0          | 0        | 0        |

**Note:**  
While filling in the flip-flop transition table, refer to the excitation table.

| Present State | Next State | FF State |   |
|---------------|------------|----------|---|
| $Q_n$         | $Q_{n+1}$  | J        | K |
| 0             | 0          | 0        | X |
| 0             | 1          | 1        | X |
| 1             | 0          | X        | 1 |
| 1             | 1          | X        | 0 |

- Step 4:

FF transition table

FF2

FF1

FF0

| Present State |       |       | Next State |          |          | JK Transition Table |       |       |       |       |       |
|---------------|-------|-------|------------|----------|----------|---------------------|-------|-------|-------|-------|-------|
| $Q_2$         | $Q_1$ | $Q_0$ | $Q_{2+}$   | $Q_{1+}$ | $Q_{0+}$ | $J_2$               | $K_2$ | $J_1$ | $K_1$ | $J_0$ | $K_0$ |
| 0             | 0     | 0     | 0          | 0        | 1        | 0                   | X     | 0     | X     | 1     | X     |
| 0             | 0     | 1     | 0          | 1        | 0        | 0                   | X     | 1     | X     | X     | 1     |
| 0             | 1     | 0     | 0          | 1        | 1        | 0                   | X     | X     | 0     | 1     | X     |
| 0             | 1     | 1     | 1          | 0        | 0        | 1                   | X     | X     | 1     | X     | 1     |
| 1             | 0     | 0     | 1          | 0        | 1        | X                   | 0     | 0     | X     | 1     | X     |
| 1             | 0     | 1     | 1          | 1        | 0        | X                   | 0     | 1     | X     | X     | 1     |
| 1             | 1     | 0     | 1          | 1        | 1        | X                   | 0     | X     | 0     | 1     | X     |
| 1             | 1     | 1     | 0          | 0        | 0        | X                   | 1     | X     | 1     | X     | 1     |

- Step 5:

Create K-map to determine the Boolean expression

| Present State |       |       | Next State |          |          | JK Transition Table |       |       |       |       |       |
|---------------|-------|-------|------------|----------|----------|---------------------|-------|-------|-------|-------|-------|
| $Q_2$         | $Q_1$ | $Q_0$ | $Q_{2+}$   | $Q_{1+}$ | $Q_{0+}$ | $J_2$               | $K_2$ | $J_1$ | $K_1$ | $J_0$ | $K_0$ |
| 0             | 0     | 0     |            |          |          | 0                   | X     | 0     | X     | 1     | X     |
| 0             | 0     | 1     |            |          |          | 0                   | X     | 1     | X     | X     | 1     |
| 0             | 1     | 0     |            |          |          | 0                   | X     | X     | 0     | 1     | X     |
| 0             | 1     | 1     |            |          |          | 1                   | X     | X     | 1     | X     | 1     |
| 1             | 0     | 0     |            |          |          | X                   | 0     | 0     | X     | 1     | X     |
| 1             | 0     | 1     |            |          |          | X                   | 0     | 1     | X     | X     | 1     |
| 1             | 1     | 0     |            |          |          | X                   | 0     | X     | 0     | 1     | X     |
| 1             | 1     | 1     |            |          |          | X                   | 1     | X     | 1     | X     | 1     |

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 | 0  | 0  | 1  | 0  |
|           | 1 | X  | X  | X  | X  |

$$J_2 = Q_1 Q_0$$

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 | 0  | 1  | X  | X  |
|           | 1 | 0  | 1  | X  | X  |

$$J_1 =$$

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 | 1  | X  | X  | 1  |
|           | 1 | 1  | X  | X  | 1  |

$$J_0 =$$

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 |    |    |    |    |
|           | 1 |    |    |    |    |

$$K_2 =$$

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 |    |    |    |    |
|           | 1 |    |    |    |    |

$$K_1 =$$

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 |    |    |    |    |
|           | 1 |    |    |    |    |

$$K_0 =$$

- Step 5:

Create K-map to determine the Boolean expression

| Present State |       |       | Next State |          |          | JK Transition Table |       |       |       |       |       |
|---------------|-------|-------|------------|----------|----------|---------------------|-------|-------|-------|-------|-------|
| $Q_2$         | $Q_1$ | $Q_0$ | $Q_{2+}$   | $Q_{1+}$ | $Q_{0+}$ | $J_2$               | $K_2$ | $J_1$ | $K_1$ | $J_0$ | $K_0$ |
| 0             | 0     | 0     |            |          |          | 0                   | X     | 0     | X     | 1     | X     |
| 0             | 0     | 1     |            |          |          | 0                   | X     | 1     | X     | X     | 1     |
| 0             | 1     | 0     |            |          |          | 0                   | X     | X     | 0     | 1     | X     |
| 0             | 1     | 1     |            |          |          | 1                   | X     | X     | 1     | X     | 1     |
| 1             | 0     | 0     |            |          |          | X                   | 0     | 0     | X     | 1     | X     |
| 1             | 0     | 1     |            |          |          | X                   | 0     | 1     | X     | X     | 1     |
| 1             | 1     | 0     |            |          |          | X                   | 0     | X     | 0     | 1     | X     |
| 1             | 1     | 1     |            |          |          | X                   | 1     | X     | 1     | X     | 1     |

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 | 0  | 0  | 1  | 0  |
|           | 1 | X  | X  | X  | X  |

$$J_2 = Q_1 Q_0$$

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 | 0  | 1  | X  | X  |
|           | 1 | 0  | 1  | X  | X  |

$$J_1 = Q_0$$

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 | 1  | X  | X  | 1  |
|           | 1 | 1  | X  | X  | 1  |

$$J_0 = 1$$

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 | X  | X  | X  | X  |
|           | 1 | 0  | 0  | 1  | 0  |

$$K_2 = Q_1 Q_0$$

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 | X  | X  | 1  | 0  |
|           | 1 | X  | X  | 1  | 0  |

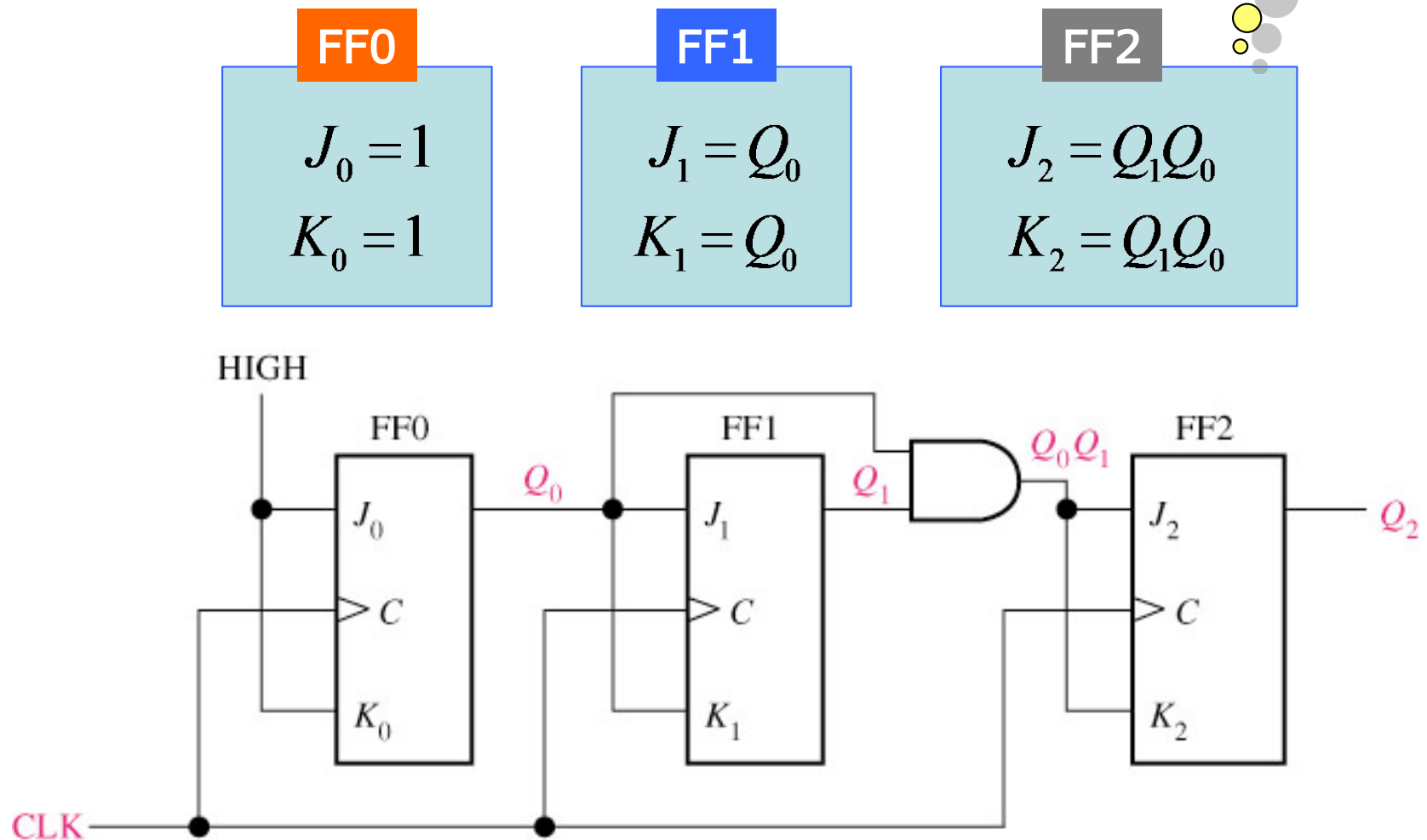
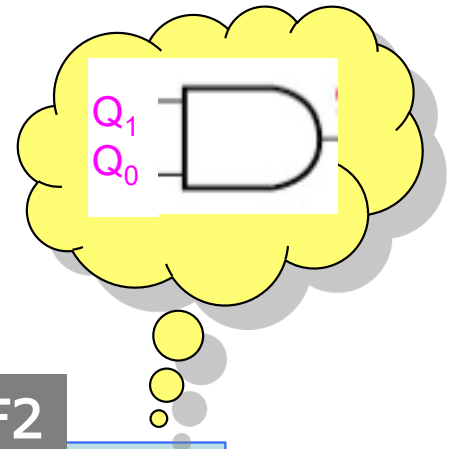
$$K_1 = Q_0$$

| $Q_1 Q_0$ |   |    |    |    |    |
|-----------|---|----|----|----|----|
| $Q_2$     |   | 00 | 01 | 11 | 10 |
|           | 0 | X  | 1  | 1  | X  |
|           | 1 | X  | 1  | 1  | X  |

$$K_0 = 1$$

- Step 6:

The implementation of 3-bit synchronous counter





Home  
work

**Exercise 8b.5:** Design 3-bit up synchronous counter that using D flip-flop with positive edge triggered. Show all steps clearly.

**Exercise 8b.6:** Design 4-bit up synchronous counter that using the following flip-flops with negative edge triggered. Show all steps clearly.

- (a) D flip-flop.
- (b) T flip-flop.



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# Synchronous Counters:

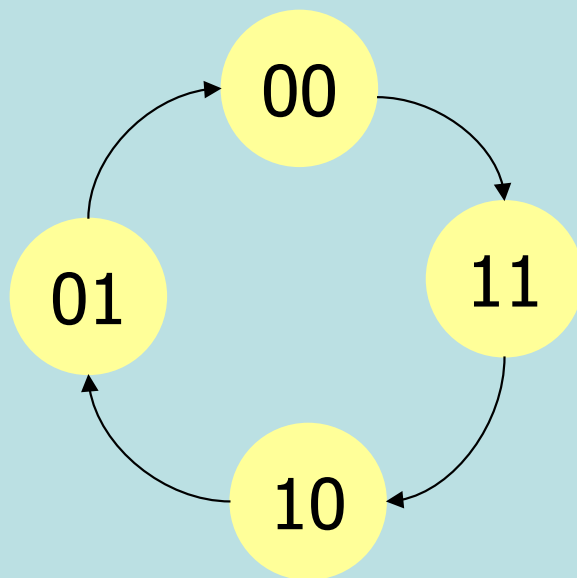
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Down

## 2-bit Down Synchronous Counter: D Flip-flop

- **Step 1:**  
To design 2-bit down synchronous counter using D flip-flop.  
There is no input or output element in this design.

- **Step 2:** Draw state diagram



- **Step 3:** Create next state table

| Present State |       | Next State |          |
|---------------|-------|------------|----------|
| $Q_1$         | $Q_0$ | $Q_{1+}$   | $Q_{0+}$ |
| 0             | 0     | 1          | 1        |
| 0             | 1     | 0          | 0        |
| 1             | 0     | 0          | 1        |
| 1             | 1     | 1          | 0        |

| $Q_{n+1}$ | D |
|-----------|---|
| 0         | 0 |
| 1         | 1 |

- Step 4:

Draw flip-flop transition table. For D flip-flop, D input **is the same** as the next state values.

| Present State |       | Next State |          | D Transition |       |
|---------------|-------|------------|----------|--------------|-------|
| $Q_1$         | $Q_0$ | $Q_{1+}$   | $Q_{0+}$ | $D_1$        | $D_0$ |
| 0             | 0     | 1          | 1        | 1            | 1     |
| 0             | 1     | 0          | 0        | 0            | 0     |
| 1             | 0     | 0          | 1        | 0            | 1     |
| 1             | 1     | 1          | 0        | 1            | 0     |

| $Q_{n+1}$ | D |
|-----------|---|
| 0         | 0 |
| 1         | 1 |

- Step 4:

Draw flip-flop transition table. For D flip-flop, D input is the same as the next state values.

| Present State |       | Next State |          | D Transition |       |
|---------------|-------|------------|----------|--------------|-------|
| $Q_1$         | $Q_0$ | $Q_{1+}$   | $Q_{0+}$ | $D_1$        | $D_0$ |
| 0             | 0     |            |          | 1            | 1     |
| 0             | 1     |            |          | 0            | 0     |
| 1             | 0     |            |          | 0            | 1     |
| 1             | 1     |            |          | 1            | 0     |

- Step 5: K-Map.

|       |   | $Q_0$ |   |
|-------|---|-------|---|
|       |   | 0     | 1 |
| $Q_1$ | 0 | 1     | 0 |
|       | 1 | 1     | 0 |

$$D_0 = \bar{Q}_0$$

|       |   | $Q_0$ |   |
|-------|---|-------|---|
|       |   | 0     | 1 |
| $Q_1$ | 0 | 1     | 0 |
|       | 1 | 0     | 1 |

$$D_1 = \bar{Q}_1 \bar{Q}_0 + Q_1 Q_0$$

$$D_1 = Q_0 \odot Q_1$$

**Exercise 8b.7:** Draw the circuit for of 2-bit down synchronous counter using D flip-flop with  $D_1 = Q_0 \odot Q_1$  and  $D_0 = \overline{Q_0}$

**Exercise 8b.8:** Design 3-bit down synchronous counter that using the following flip-flops with negative edge triggered. Show all steps clearly.

- (a) J-K flip-flop.
- (b) D flip-flop.
- (c) T flip-flop.

- Exercise 8b.9:** Design 4-bit down synchronous counter that using the following flip-flops with positive edge triggered. Show all steps clearly.
- (a) D flip-flop.
  - (b) T flip-flop.



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# Synchronous Counters:

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## Up-Down

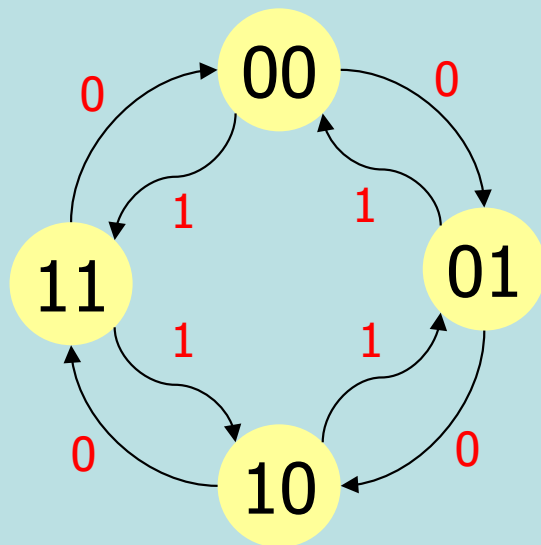
# 2-bit UP/DOWN Synchronous Counter: D Flip-flop

- Step 1:

To design 2-bit up-down synchronous counter using D flip-flop based on input X. When  $X=0 \rightarrow$  Count UP and  $X=1 \rightarrow$  Count DOWN

- Step 2:

Given the state diagram.



- Step 3: Create next state table

| Input,<br>X | Present State |        | Next State |            |
|-------------|---------------|--------|------------|------------|
|             | $Q1_n$        | $Q0_n$ | $Q1_{n+1}$ | $Q0_{n+1}$ |
| Count up    | 0             | 0      | 0          | 1          |
|             | 0             | 1      | 1          | 0          |
|             | 1             | 0      | 1          | 1          |
|             | 1             | 1      | 0          | 0          |
| Count down  | 1             | 0      | 1          | 1          |
|             | 1             | 0      | 1          | 0          |
|             | 1             | 1      | 0          | 1          |
|             | 1             | 1      | 1          | 0          |

- Step 4:  
D flip-flop transition table.

| Input,<br>X | Present State |        | Next State |            | D FF |    |
|-------------|---------------|--------|------------|------------|------|----|
|             | $Q1_n$        | $Q0_n$ | $Q1_{n+1}$ | $Q0_{n+1}$ | D1   | D0 |
| 0           | 0             | 0      | 0          | 1          | 0    | 1  |
| 0           | 0             | 1      | 1          | 0          | 1    | 0  |
| 0           | 1             | 0      | 1          | 1          | 1    | 1  |
| 0           | 1             | 1      | 0          | 0          | 0    | 0  |
| 1           | 0             | 0      | 1          | 1          | 1    | 1  |
| 1           | 0             | 1      | 0          | 0          | 0    | 0  |
| 1           | 1             | 0      | 0          | 1          | 0    | 1  |
| 1           | 1             | 1      | 1          | 0          | 1    | 0  |

a checker board pattern

|         |    |    |    |    |
|---------|----|----|----|----|
| cd \ ab | 00 | 01 | 11 | 10 |
|         | 00 | 1  |    | 1  |
| 01      | 1  |    | 1  |    |
| 11      |    | 1  |    | 1  |
| 10      | 1  |    | 1  |    |

$$X = A \oplus B \oplus C \oplus D$$

- Step 4:  
D flip-flop transition table.

| Input,<br>X | Present State |        | Next State |            | D FF |    |
|-------------|---------------|--------|------------|------------|------|----|
|             | $Q1_n$        | $Q0_n$ | $Q1_{n+1}$ | $Q0_{n+1}$ | D1   | D0 |
| 0           | 0             | 0      |            |            | 0    | 1  |
| 0           | 0             | 1      |            |            | 1    | 0  |
| 0           | 1             | 0      |            |            | 1    | 1  |
| 0           | 1             | 1      |            |            | 0    | 0  |
| 1           | 0             | 0      |            |            | 1    | 1  |
| 1           | 0             | 1      |            |            | 0    | 0  |
| 1           | 1             | 0      |            |            | 0    | 1  |
| 1           | 1             | 1      |            |            | 1    | 0  |

- Step 5:  
Define logic equation.  
(create k-maps)

|   |   |          |    |    |    |
|---|---|----------|----|----|----|
|   |   | $Q_1Q_0$ |    |    |    |
| X |   | 00       | 01 | 11 | 10 |
|   | 0 | 0        | 1  | 0  | 1  |
|   | 1 | 1        | 0  | 1  | 0  |

$$D_1 = X \oplus Q_1 \oplus Q_0$$

|   |   |          |    |    |    |
|---|---|----------|----|----|----|
|   |   | $Q_1Q_0$ |    |    |    |
| X |   | 00       | 01 | 11 | 10 |
|   | 0 | 1        | 0  | 0  | 1  |
|   | 1 | 1        | 0  | 0  | 1  |

$$D_0 = \overline{Q_0}$$

**Exercise 8b.10:** Draw the circuit for of 2-bit up-down synchronous counter using D flip-flop with:

$$D_1 = X \oplus Q_1 \oplus Q_0$$

$$D_0 = \overline{Q_0}$$



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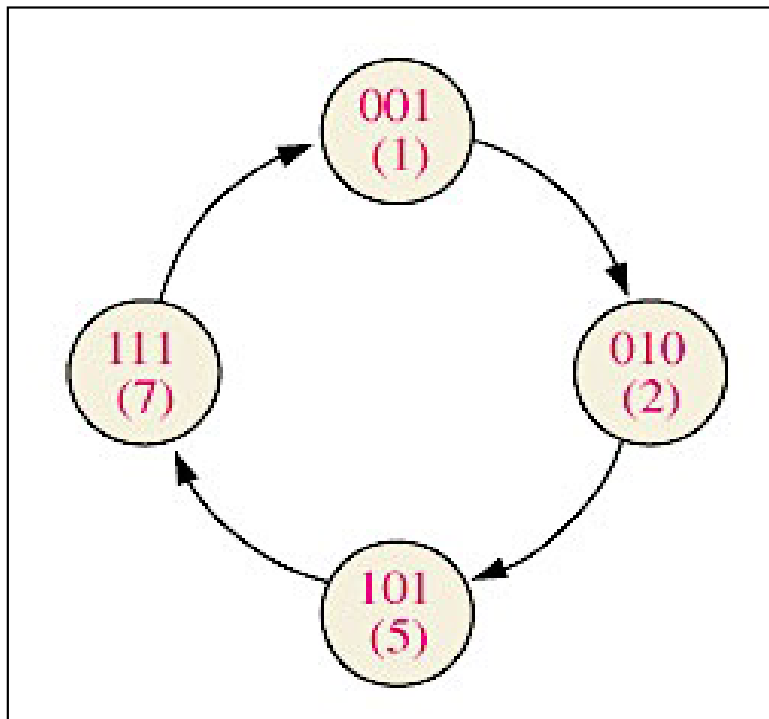
# Synchronous Counters:

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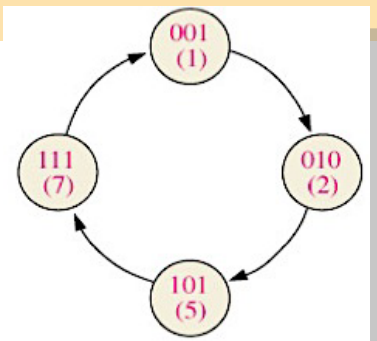
Arbitrary Sequences || BCD Decade Counter ||  
Cascaded Counter

# Counter for Arbitrary Sequences

- Design the counter base on the given state diagram using T flip-flop.



- The counter is 3-bit counter
- The total number state =  $2^3 = 8$
- Only 4 state (1, 2, 5, 7)  
→ **Valid State**
- Other states (0, 3, 4, 6)  
→ **Invalid State**  
(never occur / **don't care**)

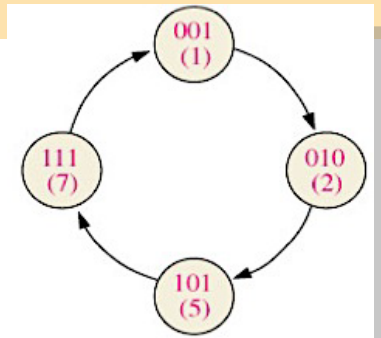


- State table and T flip-flop transition table.

| Present State |          |          | Next State |            |            | T FF  |       |       |
|---------------|----------|----------|------------|------------|------------|-------|-------|-------|
| $Q_{2n}$      | $Q_{1n}$ | $Q_{0n}$ | $Q_{2n+1}$ | $Q_{1n+1}$ | $Q_{0n+1}$ | $T_2$ | $T_1$ | $T_0$ |
| 0             | 0        | 0        | X          | X          | X          | X     | X     | X     |
| 0             | 0        | 1        |            |            |            |       |       |       |
| 0             | 1        | 0        |            |            |            |       |       |       |
| 0             | 1        | 1        | X          | X          | X          | X     | X     | X     |
| 1             | 0        | 0        | X          | X          | X          | X     | X     | X     |
| 1             | 0        | 1        |            |            |            |       |       |       |
| 1             | 1        | 0        | X          | X          | X          | X     | X     | X     |
| 1             | 1        | 1        |            |            |            |       |       |       |

\*\*Fill in the next state and T FF transition column

| $Q_{n+1}$        | T |
|------------------|---|
| $Q_n$            | 0 |
| $\overline{Q_n}$ | 1 |



- State table and T flip-flop transition table.

FF2

FF1

FF0

| Present State |          |          | Next State |            |            | T FF  |       |       |
|---------------|----------|----------|------------|------------|------------|-------|-------|-------|
| $Q_{2n}$      | $Q_{1n}$ | $Q_{0n}$ | $Q_{2n+1}$ | $Q_{1n+1}$ | $Q_{0n+1}$ | $T_2$ | $T_1$ | $T_0$ |
| 0             | 0        | 0        | X          | X          | X          | X     | X     | X     |
| 0             | 0        | 1        | 0          | 1          | 0          | 0     | 1     | 1     |
| 0             | 1        | 0        | 1          | 0          | 1          | 1     | 1     | 1     |
| 0             | 1        | 1        | X          | X          | X          | X     | X     | X     |
| 1             | 0        | 0        | X          | X          | X          | X     | X     | X     |
| 1             | 0        | 1        | 1          | 1          | 1          | 0     | 1     | 0     |
| 1             | 1        | 0        | X          | X          | X          | X     | X     | X     |
| 1             | 1        | 1        | 0          | 0          | 1          | 1     | 1     | 0     |

\*\*Fill in the next state and T FF transition column

| Present State |          |          | Next State |            |            | T FF  |       |       |
|---------------|----------|----------|------------|------------|------------|-------|-------|-------|
| $Q_{2n}$      | $Q_{1n}$ | $Q_{0n}$ | $Q_{2n+1}$ | $Q_{1n+1}$ | $Q_{0n+1}$ | $T_2$ | $T_1$ | $T_0$ |
| 0             | 0        | 0        |            |            |            | X     | X     | X     |
| 0             | 0        | 1        |            |            |            | 0     | 1     | 1     |
| 0             | 1        | 0        |            |            |            | 1     | 1     | 1     |
| 0             | 1        | 1        |            |            |            | X     | X     | X     |
| 1             | 0        | 0        |            |            |            | X     | X     | X     |
| 1             | 0        | 1        |            |            |            | 0     | 1     | 0     |
| 1             | 1        | 0        |            |            |            | X     | X     | X     |
| 1             | 1        | 1        |            |            |            | 1     | 1     | 0     |

- Step 5: Create K-map to determine the Boolean expression

|       |   |          |    |    |    |
|-------|---|----------|----|----|----|
|       |   | $Q_1Q_0$ |    |    |    |
|       |   | 00       | 01 | 11 | 10 |
| $Q_2$ | 0 | X        | 0  | X  | 1  |
|       | 1 | X        | 0  | 1  | X  |

$$T_2 = Q_1$$

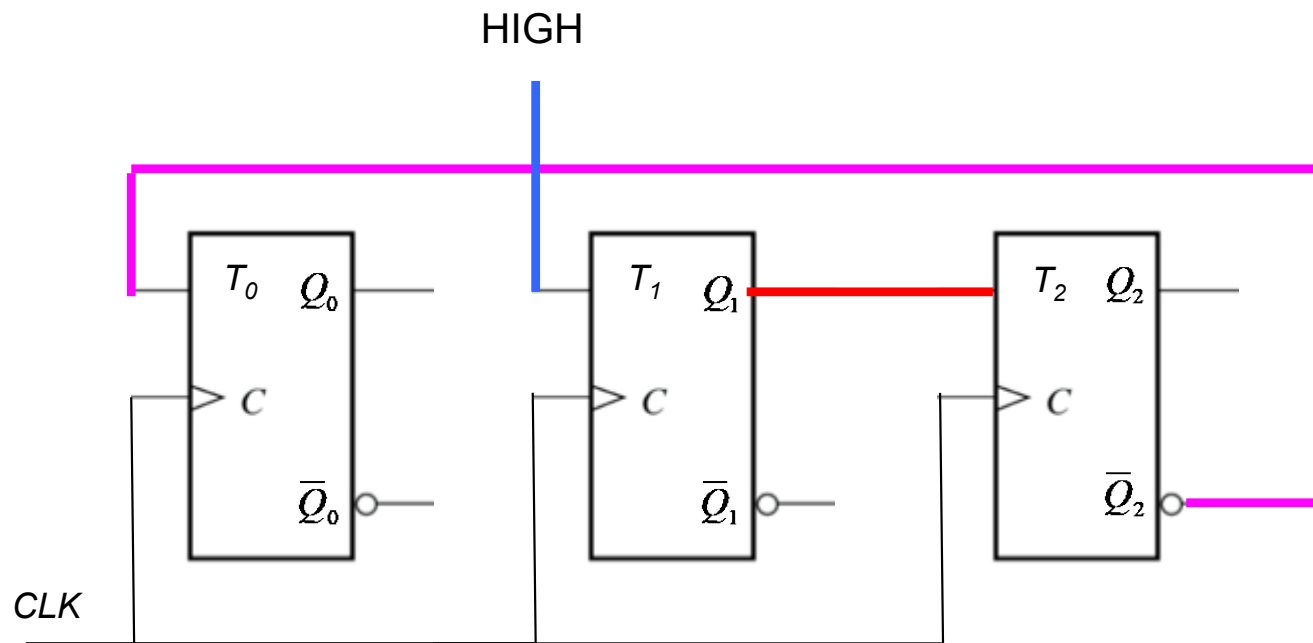
|       |   |          |    |    |    |
|-------|---|----------|----|----|----|
|       |   | $Q_1Q_0$ |    |    |    |
|       |   | 00       | 01 | 11 | 10 |
| $Q_2$ | 0 | X        | 1  | X  | 1  |
|       | 1 | X        | 1  | 1  | X  |

$$T_1 = 1$$

|       |   |          |    |    |    |
|-------|---|----------|----|----|----|
|       |   | $Q_1Q_0$ |    |    |    |
|       |   | 00       | 01 | 11 | 10 |
| $Q_2$ | 0 | X        | 1  | X  | 1  |
|       | 1 | X        | 0  | 0  | X  |

$$T_0 = \overline{Q_2}$$

- Step 6: Draw the circuit design.

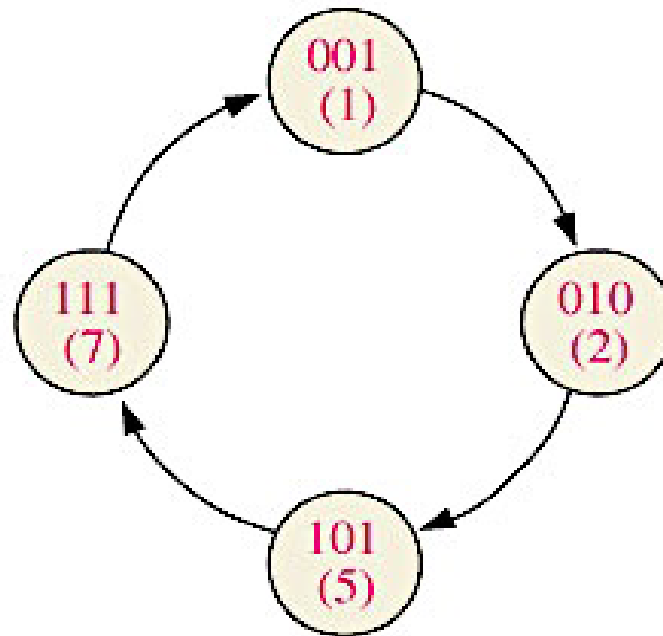


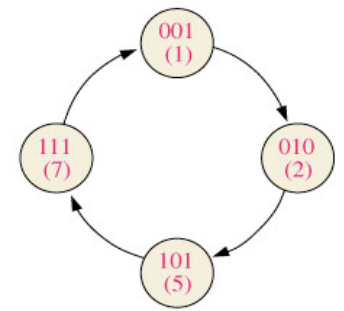
$$T_0 = \bar{Q}_2$$

$$T_1 = 1$$

$$T_2 = Q_1$$

**Exercise 8b.14:** Design a counter with the **irregular** binary count sequence as shown in the state diagram below using **J-K flip-flop**.





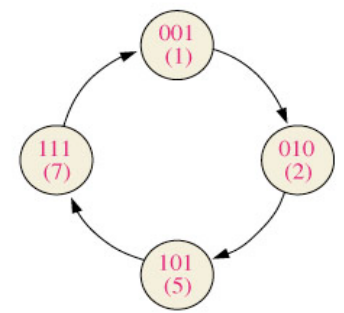
## Solution 8b.14:

- Step 3 and 4: Next state table and FF transition table.

\*\*Fill in the next state and T FF transition column

| Present State |       |       | Next State |          |          | JK Transition Table |       |       |       |       |       |
|---------------|-------|-------|------------|----------|----------|---------------------|-------|-------|-------|-------|-------|
| $Q_2$         | $Q_1$ | $Q_0$ | $Q_{2+}$   | $Q_{1+}$ | $Q_{0+}$ | $J_2$               | $K_2$ | $J_1$ | $K_1$ | $J_0$ | $K_0$ |
| 0             | 0     | 0     | X          | X        | X        |                     |       |       |       |       |       |
| 0             | 0     | 1     | 0          | 1        | 0        |                     |       |       |       |       |       |
| 0             | 1     | 0     | 1          | 0        | 1        |                     |       |       |       |       |       |
| 0             | 1     | 1     | X          | X        | X        |                     |       |       |       |       |       |
| 1             | 0     | 0     | X          | X        | X        |                     |       |       |       |       |       |
| 1             | 0     | 1     | 1          | 1        | 1        |                     |       |       |       |       |       |
| 1             | 1     | 0     | X          | X        | X        |                     |       |       |       |       |       |
| 1             | 1     | 1     | 0          | 0        | 1        |                     |       |       |       |       |       |

| Present State | Next State | FF State |   |
|---------------|------------|----------|---|
| $Q_n$         | $Q_{n+1}$  | J        | K |
| 0             | 0          | 0        | X |
| 0             | 1          | 1        | X |
| 1             | 0          | X        | 1 |
| 1             | 1          | X        | 0 |



- Step 3 and 4: Next state table and FF transition table.

FF2

FF1

FF0

| Present State |       |       | Next State |          |          | JK Transition Table |       |       |       |       |       |
|---------------|-------|-------|------------|----------|----------|---------------------|-------|-------|-------|-------|-------|
| $Q_2$         | $Q_1$ | $Q_0$ | $Q_{2+}$   | $Q_{1+}$ | $Q_{0+}$ | $J_2$               | $K_2$ | $J_1$ | $K_1$ | $J_0$ | $K_0$ |
| 0             | 0     | 0     | X          | X        | X        | X                   | X     | X     | X     | X     | X     |
| 0             | 0     | 1     | 0          | 1        | 0        | 0                   | X     | 1     | X     | X     | 1     |
| 0             | 1     | 0     | 1          | 0        | 1        | 1                   | X     | X     | 1     | 1     | X     |
| 0             | 1     | 1     | X          | X        | X        | X                   | X     | X     | X     | X     | X     |
| 1             | 0     | 0     | X          | X        | X        | X                   | X     | X     | X     | X     | X     |
| 1             | 0     | 1     | 1          | 1        | 1        | X                   | 0     | 1     | X     | X     | 0     |
| 1             | 1     | 0     | X          | X        | X        | X                   | X     | X     | X     | X     | X     |
| 1             | 1     | 1     | 0          | 0        | 1        | X                   | 1     | X     | 1     | X     | 0     |

- Step 5:

Create K-map to determine the Boolean expression

|    |   | Q1Q0 |    |    |    |
|----|---|------|----|----|----|
|    |   | 00   | 01 | 11 | 10 |
| Q2 | 0 | X    | 0  | X  | 1  |
|    | 1 | X    | X  | X  | X  |

$$J_2 = \overline{Q_0}$$

|    |   | Q1Q0 |    |    |    |
|----|---|------|----|----|----|
|    |   | 00   | 01 | 11 | 10 |
| Q2 | 0 | X    | 1  | X  | X  |
|    | 1 | X    | 1  | X  | X  |

$$J_1 = 1$$

|    |   | Q1Q0 |    |    |    |
|----|---|------|----|----|----|
|    |   | 00   | 01 | 11 | 10 |
| Q2 | 0 | X    | X  | X  | 1  |
|    | 1 | X    | X  | X  | X  |

$$J_0 = 1$$

|    |   | Q1Q0 |    |    |    |
|----|---|------|----|----|----|
|    |   | 00   | 01 | 11 | 10 |
| Q2 | 0 | X    | X  | X  | X  |
|    | 1 | X    | 0  | 1  | X  |

$$K_2 = Q_1$$

|    |   | Q1Q0 |    |    |    |
|----|---|------|----|----|----|
|    |   | 00   | 01 | 11 | 10 |
| Q2 | 0 | X    | X  | X  | 1  |
|    | 1 | X    | X  | 1  | X  |

$$K_1 = 1$$

|    |   | Q1Q0 |    |    |    |
|----|---|------|----|----|----|
|    |   | 00   | 01 | 11 | 10 |
| Q2 | 0 | X    | 1  | X  | X  |
|    | 1 | X    | 0  | 0  | X  |

$$K_0 = \overline{Q_2}$$

Step 6: Draw the circuit design

$$J_0 = 1$$

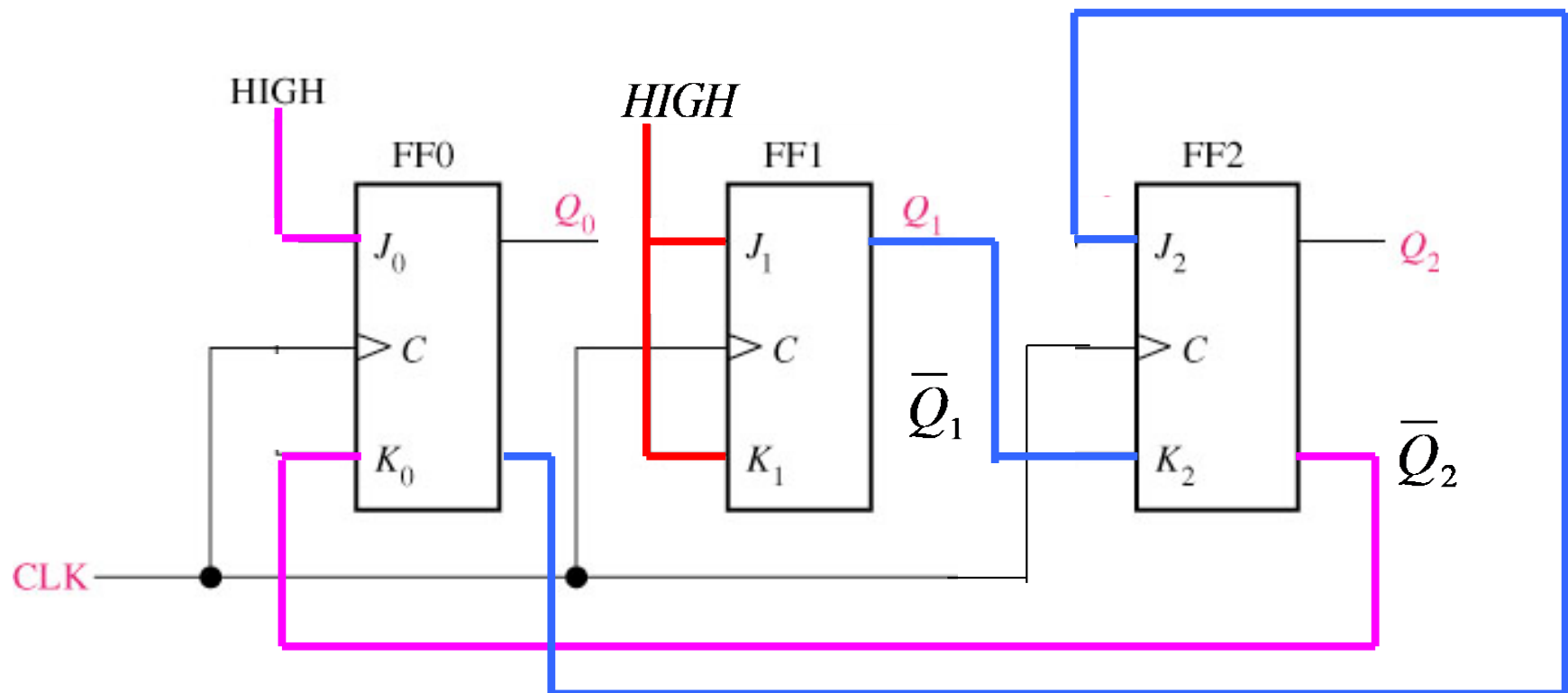
$$J_1 = 1$$

$$J_2 = \overline{Q_0}$$

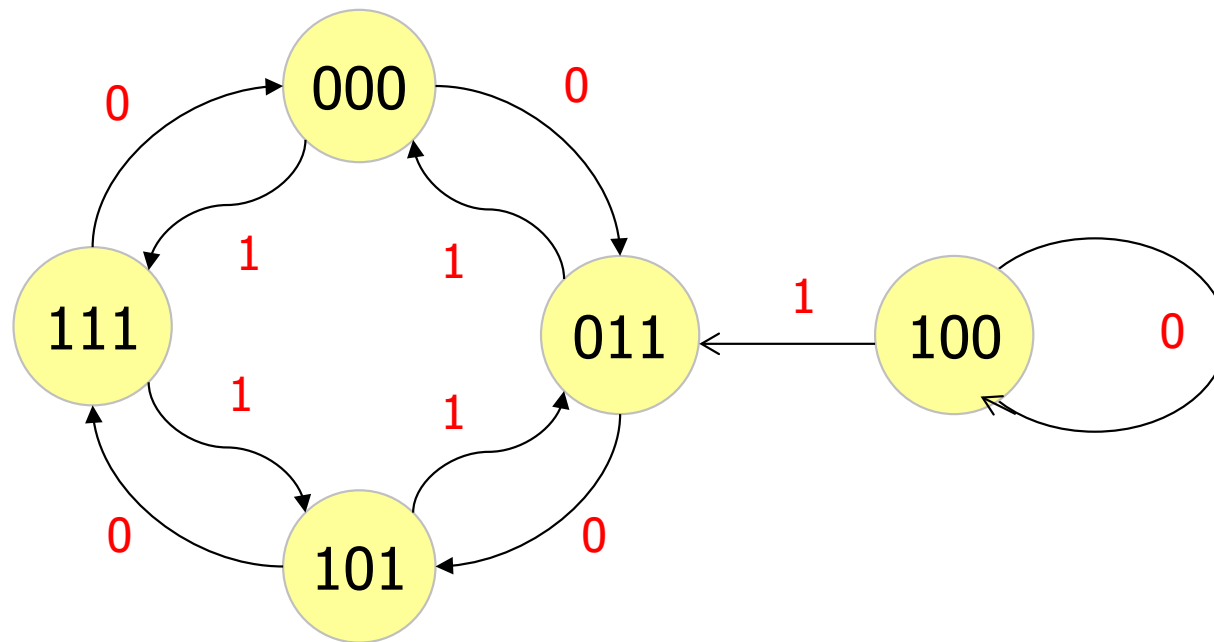
$$K_0 = \overline{Q_2}$$

$$K_1 = 1$$

$$K_2 = Q_1$$



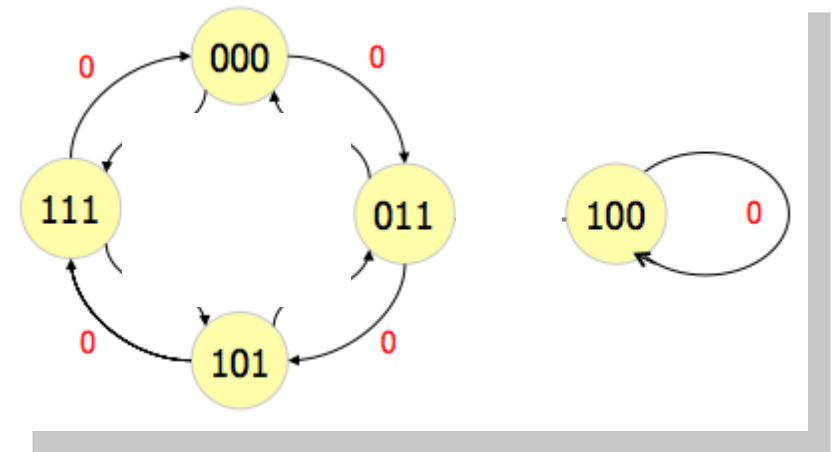
**Exercise 8b.16:** Design a synchronous counter with the binary count sequence as shown in the state diagram below using J-K flip-flop.



## Solutions:

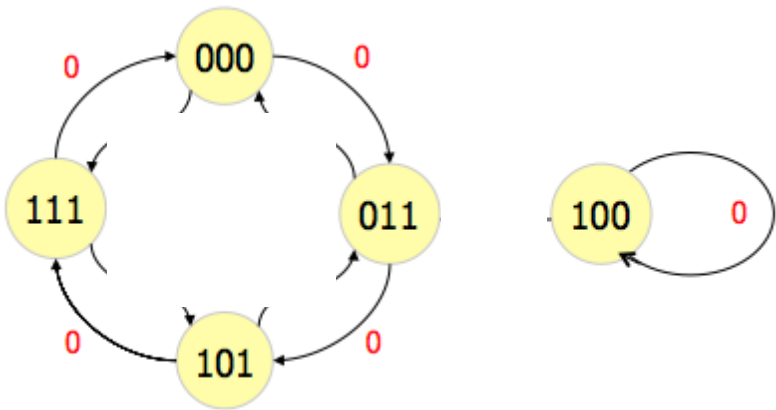
Input,  $X = 0$

| Input | Present State |       |       | Next State |          |          |
|-------|---------------|-------|-------|------------|----------|----------|
| X     | $Q_2$         | $Q_1$ | $Q_0$ | $Q_{2+}$   | $Q_{1+}$ | $Q_{0+}$ |
| 0     | 0             | 0     | 0     |            |          |          |
| 0     | 0             | 0     | 1     |            |          |          |
| 0     | 0             | 1     | 0     |            |          |          |
| 0     | 0             | 1     | 1     |            |          |          |
| 0     | 1             | 0     | 0     |            |          |          |
| 0     | 1             | 0     | 1     |            |          |          |
| 0     | 1             | 1     | 0     |            |          |          |
| 0     | 1             | 1     | 1     |            |          |          |



Solutions:

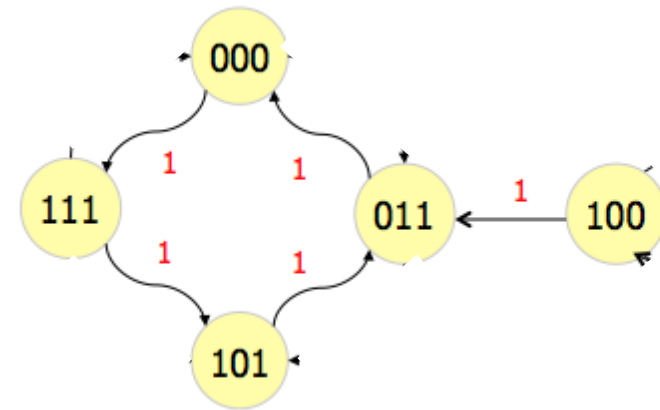
| Present State | Next State | FF State |   |
|---------------|------------|----------|---|
| $Q_n$         | $Q_{n+1}$  | J        | K |
| 0             | 0          | 0        | X |
| 0             | 1          | 1        | X |
| 1             | 0          | X        | 1 |
| 1             | 1          | X        | 0 |



Input,  $X = 0$

| Input | Present State |       |       | Next State |          |          |
|-------|---------------|-------|-------|------------|----------|----------|
| X     | $Q_2$         | $Q_1$ | $Q_0$ | $Q_{2+}$   | $Q_{1+}$ | $Q_{0+}$ |
| 0     | 0             | 0     | 0     | 0          | 1        | 1        |
| 0     | 0             | 0     | 1     | X          | X        | X        |
| 0     | 0             | 1     | 0     | X          | X        | X        |
| 0     | 0             | 1     | 1     | 1          | 0        | 1        |
| 0     | 1             | 0     | 0     | 1          | 0        | 0        |
| 0     | 1             | 0     | 1     | 1          | 1        | 1        |
| 0     | 1             | 1     | 0     | X          | X        | X        |
| 0     | 1             | 1     | 1     | 0          | 0        | 0        |

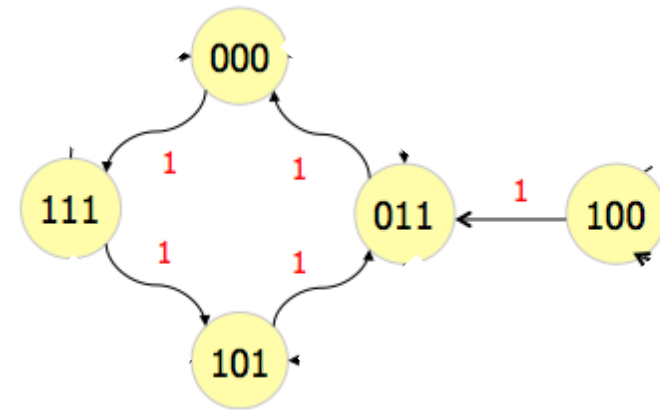
| JK Transition Table |       |       |       |       |       |
|---------------------|-------|-------|-------|-------|-------|
| $J_2$               | $K_2$ | $J_1$ | $K_1$ | $J_0$ | $K_0$ |
| 0                   | X     | 1     | X     | 1     | X     |
| X                   | X     | X     | X     | X     | X     |
| X                   | X     | X     | X     | X     | X     |
| 1                   | X     | X     | 1     | X     | 0     |
| X                   | 0     | 0     | X     | 0     | X     |
| X                   | 0     | 1     | X     | X     | 0     |
| X                   | X     | X     | X     | X     | X     |
| X                   | 1     | X     | 1     | X     | 1     |



Input,  $X = 1$

| Input | Present State |       |       | Next State |          |          |
|-------|---------------|-------|-------|------------|----------|----------|
| X     | $Q_2$         | $Q_1$ | $Q_0$ | $Q_{2+}$   | $Q_{1+}$ | $Q_{0+}$ |
| 1     | 0             | 0     | 0     |            |          |          |
| 1     | 0             | 0     | 1     |            |          |          |
| 1     | 0             | 1     | 0     |            |          |          |
| 1     | 0             | 1     | 1     |            |          |          |
| 1     | 1             | 0     | 0     |            |          |          |
| 1     | 1             | 0     | 1     |            |          |          |
| 1     | 1             | 1     | 0     |            |          |          |
| 1     | 1             | 1     | 1     |            |          |          |

| Present State | Next State | FF State |   |
|---------------|------------|----------|---|
| $Q_n$         | $Q_{n+1}$  | J        | K |
| 0             | 0          | 0        | X |
| 0             | 1          | 1        | X |
| 1             | 0          | X        | 1 |
| 1             | 1          | X        | 0 |



Input,  $X = 1$

| Input | Present State |       |       | Next State |          |          |
|-------|---------------|-------|-------|------------|----------|----------|
| X     | $Q_2$         | $Q_1$ | $Q_0$ | $Q_{2+}$   | $Q_{1+}$ | $Q_{0+}$ |
| 1     | 0             | 0     | 0     | 1          | 1        | 1        |
| 1     | 0             | 0     | 1     | X          | X        | X        |
| 1     | 0             | 1     | 0     | X          | X        | X        |
| 1     | 0             | 1     | 1     | 0          | 0        | 0        |
| 1     | 1             | 0     | 0     | 0          | 1        | 1        |
| 1     | 1             | 0     | 1     | 0          | 1        | 1        |
| 1     | 1             | 1     | 0     | X          | X        | X        |
| 1     | 1             | 1     | 1     | 1          | 0        | 1        |

| JK Transition Table |       |       |       |       |       |
|---------------------|-------|-------|-------|-------|-------|
| $J_2$               | $K_2$ | $J_1$ | $K_1$ | $J_0$ | $K_0$ |
| 1                   | X     | 1     | X     | 1     | X     |
| X                   | X     | X     | X     | X     | X     |
| X                   | X     | X     | X     | X     | X     |
| 0                   | X     | X     | 0     | X     | 0     |
| X                   | 1     | 1     | X     | 1     | X     |
| X                   | 1     | 1     | X     | X     | 0     |
| X                   | X     | X     | X     | X     | X     |
| X                   | 0     | X     | 1     | X     | 0     |

Input,  $X = 0$

| Input | Present State |       |       |
|-------|---------------|-------|-------|
| X     | $Q_2$         | $Q_1$ | $Q_0$ |
| 0     | 0             | 0     | 0     |
| 0     | 0             | 0     | 1     |
| 0     | 0             | 1     | 0     |
| 0     | 0             | 1     | 1     |
| 0     | 1             | 0     | 0     |
| 0     | 1             | 0     | 1     |
| 0     | 1             | 1     | 0     |
| 0     | 1             | 1     | 1     |

Use K-maps to derive the logic equations for each J and K.

| JK Transition Table |       |       |       |       |       |
|---------------------|-------|-------|-------|-------|-------|
| $J_2$               | $K_2$ | $J_1$ | $K_1$ | $J_0$ | $K_0$ |
| 0                   | X     | 1     | X     | 1     | X     |
| X                   | X     | X     | X     | X     | X     |
| X                   | X     | X     | X     | X     | X     |
| 1                   | X     | X     | 1     | X     | 0     |
| X                   | 0     | 0     | X     | 0     | X     |
| X                   | 0     | 1     | X     | X     | 0     |
| X                   | X     | X     | X     | X     | X     |
| X                   | 1     | X     | 1     | X     | 1     |

Input,  $X = 1$

| Input | Present State |       |       |
|-------|---------------|-------|-------|
| X     | $Q_2$         | $Q_1$ | $Q_0$ |
| 1     | 0             | 0     | 0     |
| 1     | 0             | 0     | 1     |
| 1     | 0             | 1     | 0     |
| 1     | 0             | 1     | 1     |
| 1     | 1             | 0     | 0     |
| 1     | 1             | 0     | 1     |
| 1     | 1             | 1     | 0     |
| 1     | 1             | 1     | 1     |

Use K-maps to derive the logic equations for each J and K.

| JK Transition Table |       |       |       |       |       |
|---------------------|-------|-------|-------|-------|-------|
| $J_2$               | $K_2$ | $J_1$ | $K_1$ | $J_0$ | $K_0$ |
| 1                   | X     | 1     | X     | 1     | X     |
| X                   | X     | X     | X     | X     | X     |
| X                   | X     | X     | X     | X     | X     |
| 0                   | X     | X     | 0     | X     | 0     |
| X                   | 1     | 1     | X     | 1     | X     |
| X                   | 1     | 1     | X     | X     | 0     |
| X                   | X     | X     | X     | X     | X     |
| X                   | 0     | X     | 1     | X     | 0     |

# K-maps:

\*\*Fill in the K-maps

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | 0        | X  | 1  | X  |
|        | 01 | X        | X  | X  | X  |
|        | 11 | X        | X  | X  | X  |
|        | 10 | 1        | X  | 0  | X  |

$J_2 =$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | X        | X  | X  | X  |
|        | 01 | 0        | 0  | 1  | X  |
|        | 11 | 1        | 1  | 0  | X  |
|        | 10 | X        | X  | X  | X  |

$K_2 =$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | 1        | X  | X  | X  |
|        | 01 | 0        | 1  | X  | X  |
|        | 11 | 1        | 1  | X  | X  |
|        | 10 | 1        | X  | X  | X  |

$J_1 =$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | X        | X  | 1  | X  |
|        | 01 | X        | X  | 1  | X  |
|        | 11 | X        | X  | 1  | X  |
|        | 10 | X        | X  | 0  | X  |

$K_1 =$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | 1        | X  | X  | X  |
|        | 01 | 0        | X  | X  | X  |
|        | 11 | 1        | X  | X  | X  |
|        | 10 | 1        | X  | X  | X  |

$J_0 =$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | X        | X  | 0  | X  |
|        | 01 | X        | 0  | 1  | X  |
|        | 11 | X        | 0  | 0  | X  |
|        | 10 | X        | X  | 0  | X  |

$K_0 =$

# K-maps:

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | 0        | X  | 1  | X  |
|        | 01 | X        | X  | X  | X  |
|        | 11 | X        | X  | X  | X  |
|        | 10 | 1        | X  | 0  | X  |

$J_2 =$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | X        | X  | X  | X  |
|        | 01 | 0        | 0  | 1  | X  |
|        | 11 | 1        | 1  | 0  | X  |
|        | 10 | X        | X  | X  | X  |

$K_2 =$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | 1        | X  | X  | X  |
|        | 01 | 0        | 1  | X  | X  |
|        | 11 | 1        | 1  | X  | X  |
|        | 10 | 1        | X  | X  | X  |

$J_1 =$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | X        | X  | 1  | X  |
|        | 01 | X        | X  | 1  | X  |
|        | 11 | X        | X  | 1  | X  |
|        | 10 | X        | X  | 0  | X  |

$K_1 =$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | 1        | X  | X  | X  |
|        | 01 | 0        | X  | X  | X  |
|        | 11 | 1        | X  | X  | X  |
|        | 10 | 1        | X  | X  | X  |

$J_0 =$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $xQ_2$ | 00 | X        | X  | 0  | X  |
|        | 01 | X        | 0  | 1  | X  |
|        | 11 | X        | 0  | 0  | X  |
|        | 10 | X        | X  | 0  | X  |

$K_0 =$

# K-maps:

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $XQ_2$ | 00 | 0        | X  | 1  | X  |
|        | 01 | X        | X  | X  | X  |
|        | 11 | X        | X  | X  | X  |
|        | 10 | 1        | X  | 0  | X  |

$$J_2 = X\bar{Q}_1 + \bar{X}Q_1$$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $XQ_2$ | 00 | X        | X  | X  | X  |
|        | 01 | 0        | 0  | 1  | X  |
|        | 11 | 1        | 1  | 0  | X  |
|        | 10 | X        | X  | X  | X  |

$$K_2 = X\bar{Q}_1 + \bar{X}Q_1$$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $XQ_2$ | 00 | 1        | X  | X  | X  |
|        | 01 | 0        | 1  | X  | X  |
|        | 11 | 1        | 1  | X  | X  |
|        | 10 | 1        | X  | X  | X  |

$$J_1 = X + \bar{Q}_2 + Q_0$$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $XQ_2$ | 00 | X        | X  | 1  | X  |
|        | 01 | X        | X  | 1  | X  |
|        | 11 | X        | X  | 1  | X  |
|        | 10 | X        | X  | 0  | X  |

$$K_1 = \bar{X} + Q_2$$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $XQ_2$ | 00 | 1        | X  | X  | X  |
|        | 01 | 0        | X  | X  | X  |
|        | 11 | 1        | X  | X  | X  |
|        | 10 | 1        | X  | X  | X  |

$$J_0 = X + \bar{Q}_2$$

|        |    | $Q_1Q_0$ |    |    |    |
|--------|----|----------|----|----|----|
|        |    | 00       | 01 | 11 | 10 |
| $XQ_2$ | 00 | X        | X  | 0  | X  |
|        | 01 | X        | 0  | 1  | X  |
|        | 11 | X        | 0  | 0  | X  |
|        | 10 | X        | X  | 0  | X  |

$$K_0 = \bar{X}Q_2Q_1$$

Draw the counter design.

FF0

$$J_0 = X + \bar{Q}_2$$

$$K_0 = \bar{X}Q_2Q_1$$

FF1

$$J_1 = X + \bar{Q}_2 + Q_0$$

$$K_1 = \bar{X} + Q_2$$

FF2

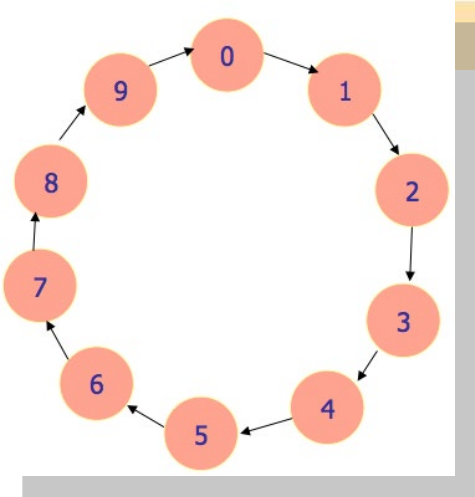
$$J_2 = X\bar{Q}_1 + \bar{X}Q_1$$

$$K_2 = X\bar{Q}_1 + \bar{X}Q_1$$

$$J_2 = K_2 = X \oplus Q_1$$

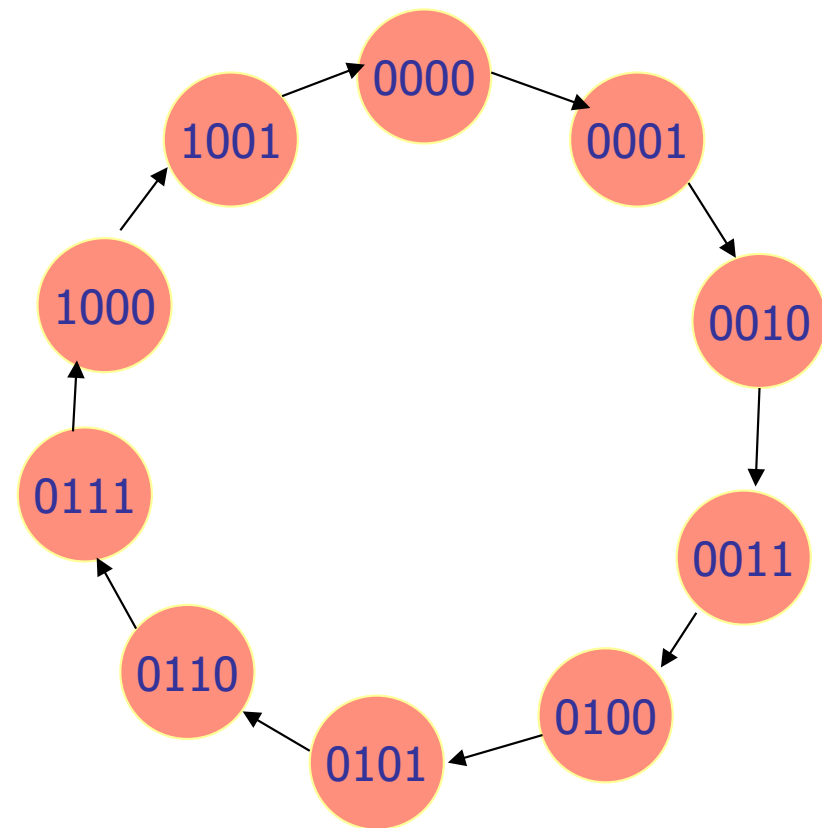
(XOR)





# Synchronous BCD Decade Counter

- Synchronous decade counter counts from 0 to 9 and then recycles to 0 again.
- 4 flip-flops are required, and the unused states (10 to 15) are taken as **don't care** terms.



**Exercise 8b.17:** Draw the design a synchronous BCD decade counter using the T flip-flop.

\*\*Fill in the  
T0 column

| Present State |       |       |       | Next State |          |          |          | T FF Transition |       |       |       |
|---------------|-------|-------|-------|------------|----------|----------|----------|-----------------|-------|-------|-------|
| $Q_3$         | $Q_2$ | $Q_1$ | $Q_0$ | $Q_{3+}$   | $Q_{2+}$ | $Q_{1+}$ | $Q_{0+}$ | $T_3$           | $T_2$ | $T_1$ | $T_0$ |
| 0             | 0     | 0     | 0     | 0          | 0        | 0        | 1        | 0               | 0     | 0     | 1     |
| 0             | 0     | 0     | 1     | 0          | 0        | 1        | 0        | 0               | 0     | 1     | 1     |
| 0             | 0     | 1     | 0     | 0          | 0        | 1        | 1        | 0               | 0     | 0     | 1     |
| 0             | 0     | 1     | 1     | 0          | 1        | 0        | 0        | 0               | 1     | 1     | 1     |
| 0             | 1     | 0     | 0     | 0          | 1        | 0        | 1        | 0               | 0     | 0     | 1     |
| 0             | 1     | 0     | 1     | 0          | 1        | 1        | 0        | 0               | 0     | 1     | 1     |
| 0             | 1     | 1     | 0     | 0          | 1        | 1        | 1        | 0               | 0     | 0     | 1     |
| 0             | 1     | 1     | 1     | 1          | 0        | 0        | 0        | 1               | 1     | 1     | 1     |
| 1             | 0     | 0     | 0     | 1          | 0        | 0        | 1        | 0               | 0     | 0     | 1     |
| 1             | 0     | 0     | 1     | 0          | 0        | 0        | 0        | 1               | 0     | 0     | 1     |
| 1             | 0     | 1     | 0     | X          | X        | X        | X        | X               | X     | X     | X     |
| 1             | 0     | 1     | 1     | X          | X        | X        | X        | X               | X     | X     | X     |
| 1             | 1     | 0     | 0     | X          | X        | X        | X        | X               | X     | X     | X     |
| 1             | 1     | 0     | 1     | X          | X        | X        | X        | X               | X     | X     | X     |
| 1             | 1     | 1     | 0     | X          | X        | X        | X        | X               | X     | X     | X     |
| 1             | 1     | 1     | 1     | X          | X        | X        | X        | X               | X     | X     | X     |

# Home work

\*\*Fill in the K-maps

Self-Test:

Fill in the K-map to simplify the equations.

|          |    | $Q_1Q_0$ |    |    |    |
|----------|----|----------|----|----|----|
|          |    | 00       | 01 | 11 | 10 |
| $Q_3Q_2$ | 00 |          |    |    |    |
|          | 01 |          |    |    |    |
|          | 11 |          |    |    |    |
|          | 10 |          |    |    |    |

|          |    | $Q_1Q_0$ |    |    |    |
|----------|----|----------|----|----|----|
|          |    | 00       | 01 | 11 | 10 |
| $Q_3Q_2$ | 00 |          |    |    |    |
|          | 01 |          |    |    |    |
|          | 11 |          |    |    |    |
|          | 10 |          |    |    |    |

$T_3 =$

|          |    | $Q_1Q_0$ |    |    |    |
|----------|----|----------|----|----|----|
|          |    | 00       | 01 | 11 | 10 |
| $Q_3Q_2$ | 00 |          |    |    |    |
|          | 01 |          |    |    |    |
|          | 11 |          |    |    |    |
|          | 10 |          |    |    |    |

$T_2 =$

|          |    | $Q_1Q_0$ |    |    |    |
|----------|----|----------|----|----|----|
|          |    | 00       | 01 | 11 | 10 |
| $Q_3Q_2$ | 00 |          |    |    |    |
|          | 01 |          |    |    |    |
|          | 11 |          |    |    |    |
|          | 10 |          |    |    |    |

$T_1 =$

$T_0 =$

**Exercise 8b.18:** Design a synchronous BCD decade counter using the **D flip-flop**.



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# Cascaded Counter

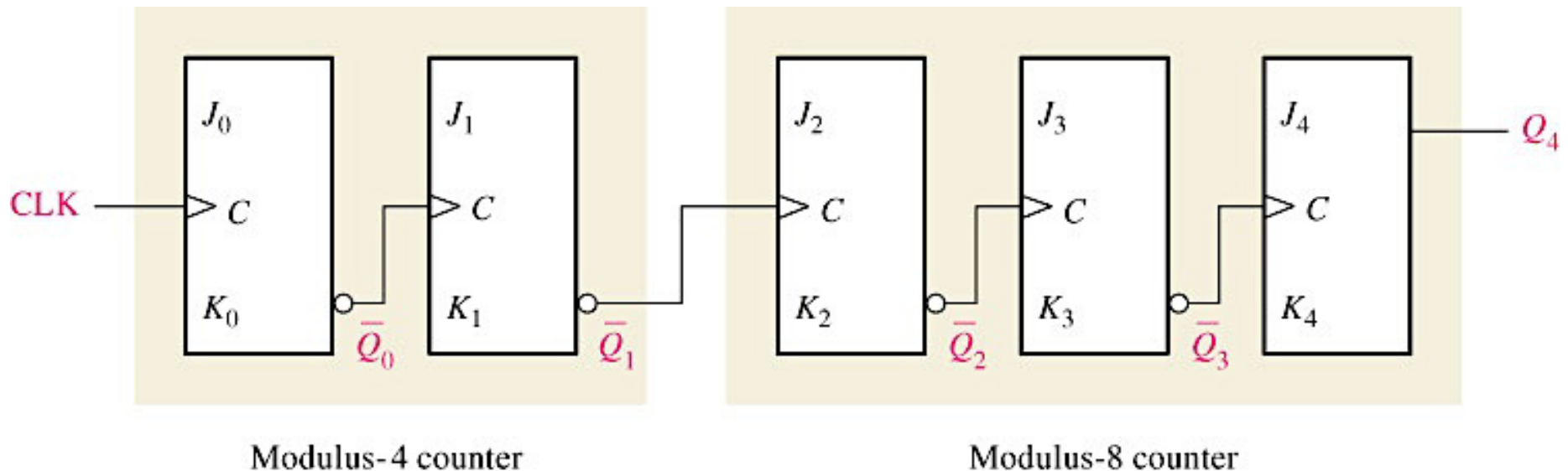
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# Cascaded Counter

- Counter can be connected to achieve higher modulus operation.
- Cascading means that the last-stage output of one counter drives the input of the next counter.

## Example:

- Two counters, modulus-4 and modulus-8 connected in cascade, can achieve count until 32 CLK (modulus-32). (i.e  $4 \times 8$ )

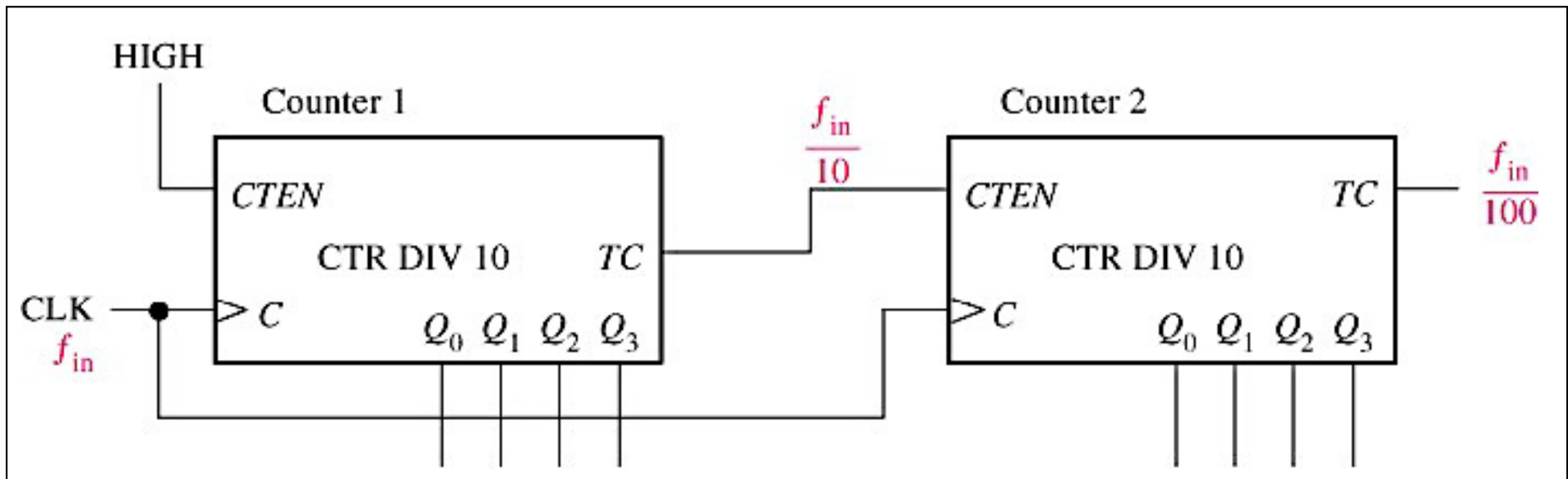


## Recap: Modulus

- The **modulus** → **number of unique states** through which the counter will sequence.
- Maximum possible number of states =  $2^N$ ,  
 $N$  is the number of flip-flops in the counter.
- Example : **Modulus 8 =  $2^3$**  (Need **3 flip flops**)
- Counter can be designed to have **less number of states** in their sequence → **Truncated sequence**.
  - One common modulus for counters → ten (Modulus 10).
  - It called BCD decade counters (as explained in previous slides).

# Cascaded Counter: Modulus-100 Counter

- Modulus-100 counter using 2 cascaded decade counters.
- This counter can be viewed as a frequency divider.
- It divides the input clock frequency by 100.

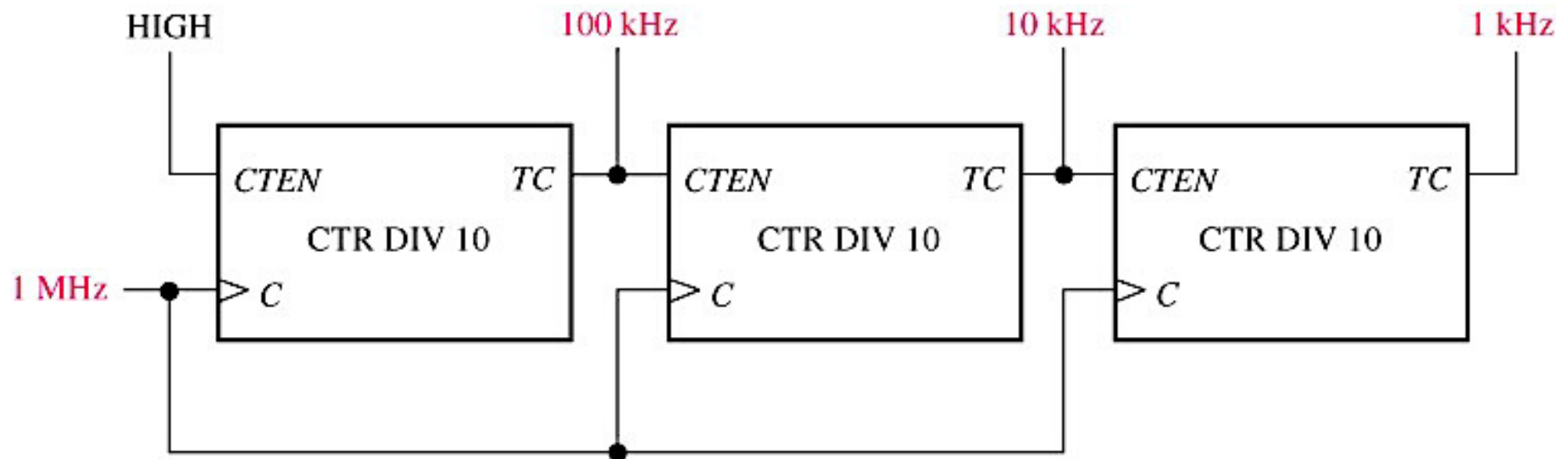


**CTEN** – Count Enable function  
**TC** – Terminal Count function

(Total MOD = 10 x 10 = 100)

## Cascaded Counter: Modulus-1000 Counter

- Three cascaded decade counters forming a divide-by-1000 frequency divider.



$$\begin{aligned}\text{Total MOD} &= 10 \times 10 \times 10 \\ &= 1000\end{aligned}$$

Basis clock frequency of 1 MHz and you wish to obtain 100kHz, 10Hz, and 1kHz, a series of cascaded decade counters can be used.

- If 1 MHz signal is divided by 10, the output is 100kHz.
- Then if the 100 kHz signal is divided by 10, the output is 10kHz.
- Further division by 10 gives the 1 kHz frequency.

**Exercise 8b.19:** Two type of counters, modulus-4 and modulus-8 need to be used to achieve count up to modulus- $n$  ( $n$  CLK).

- a) How to cascade the counters to achieve count until 32 CLK (modulus-32)?
- b) What is the frequency produced by each counter given an initial frequency is 800MHz?

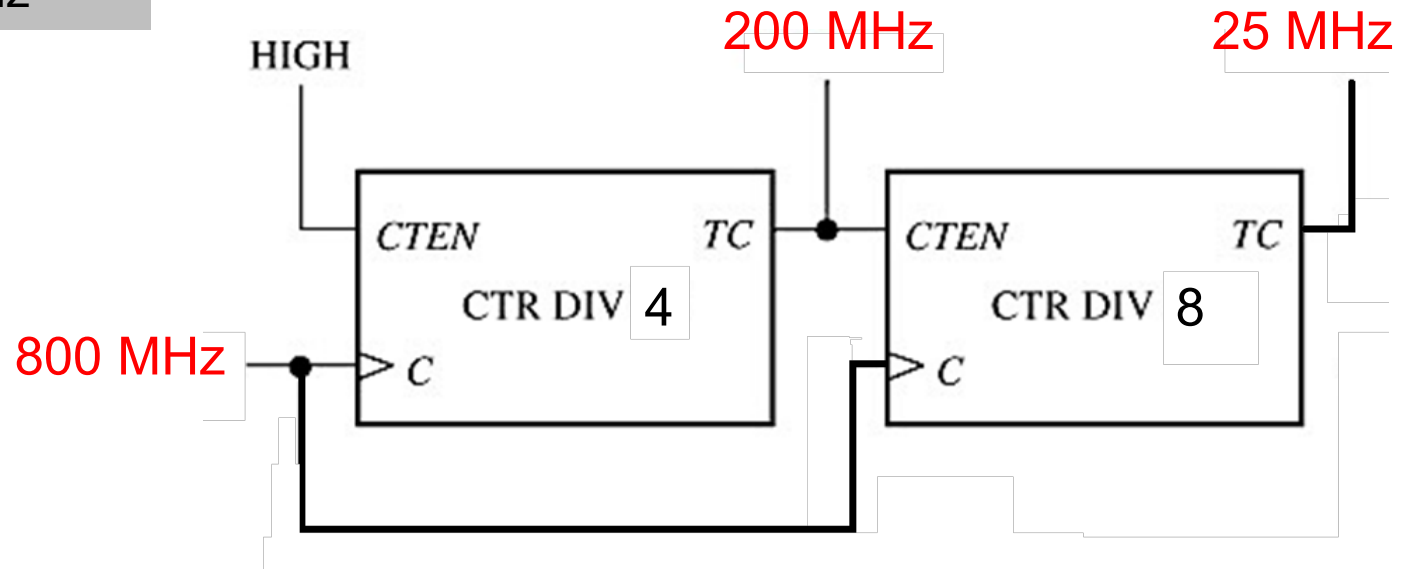
**Solutions:**

- a) Modulus-4 x Modulus-8, or  
Modulus-8 x Modulus-4.

b)

Modulus-4:  
 $= 800\text{MHz} / 4$   
 $= 200\text{MHz}$

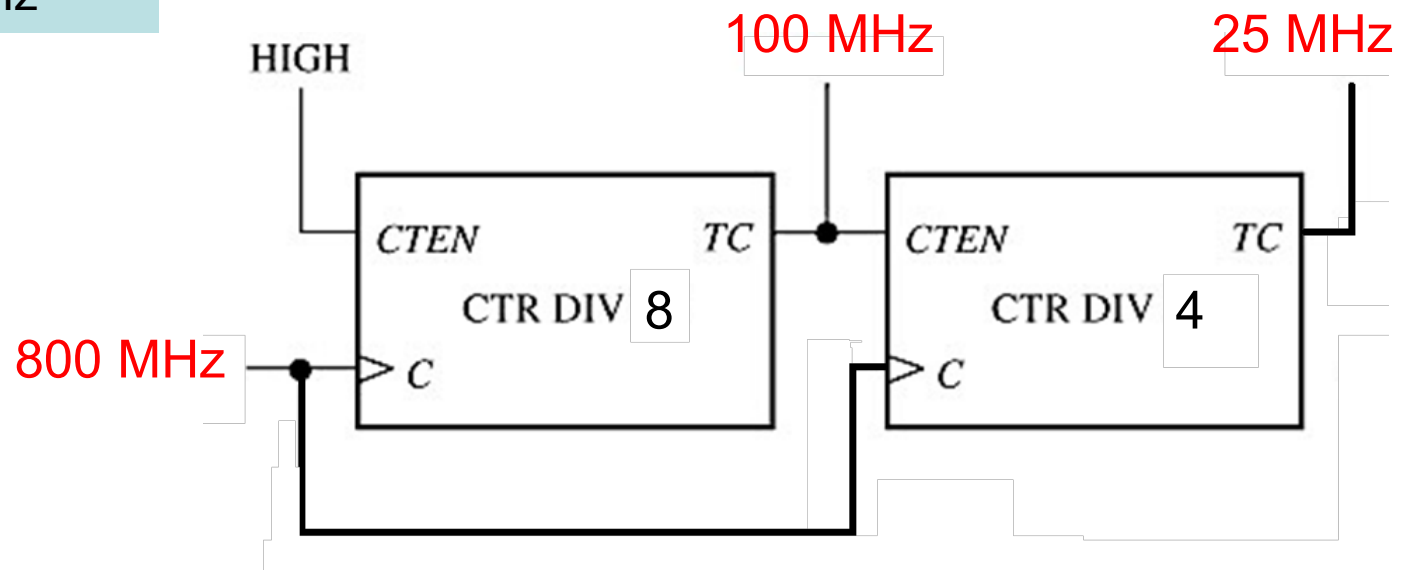
Modulus-8:  
 $= 200\text{MHz} / 8$   
 $= 25\text{MHz}$



b)

Modulus-8:  
 $= 800\text{MHz} / 8$   
 $= 100\text{MHz}$

Modulus-4:  
 $= 100\text{MHz} / 4$   
 $= 25\text{MHz}$





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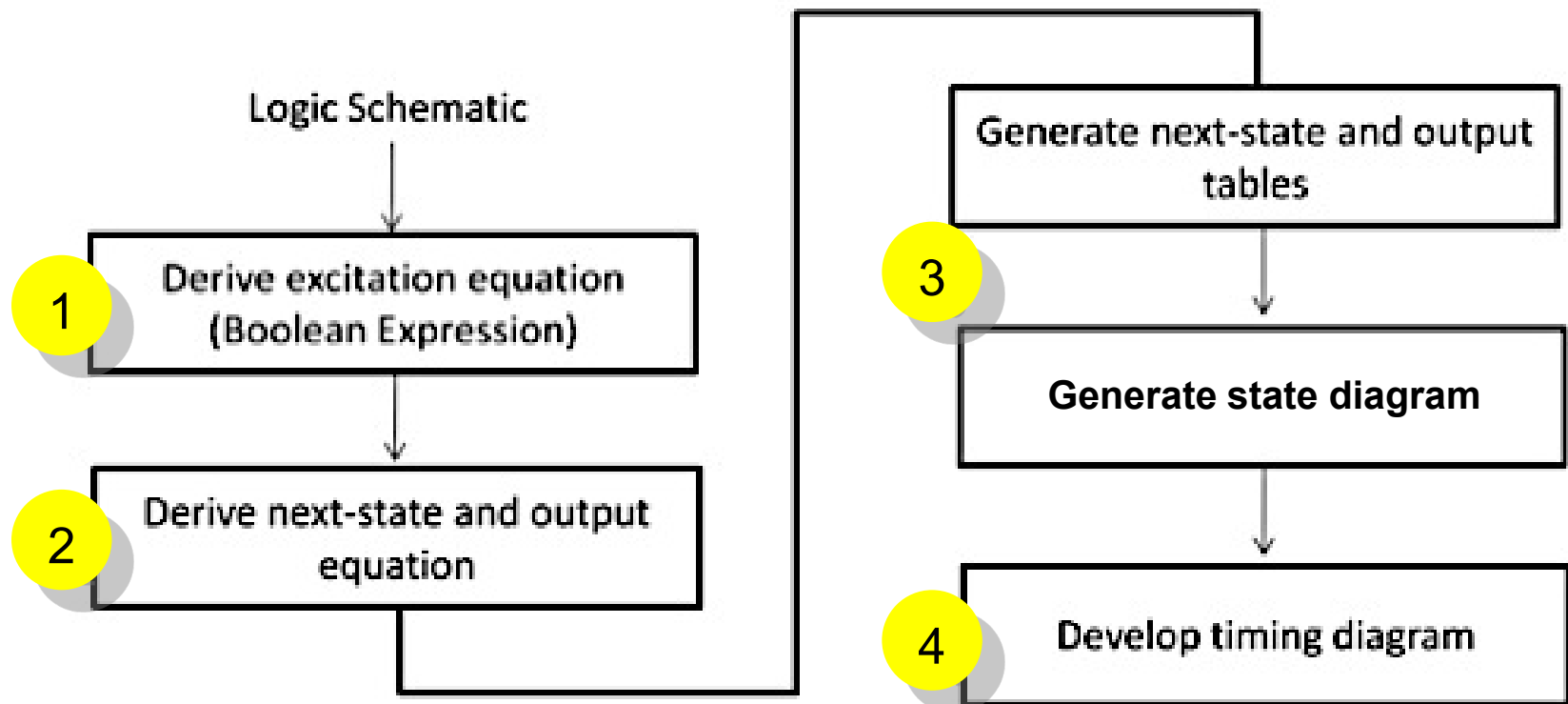
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# Analysis of Sequential Circuits

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# Analysis of Sequential Circuits

- Behavior of a sequential circuit is determined from the **inputs**, the **outputs** and the **states** of its flip-flops.
- Both the **output and the next state** are a function of the **inputs and the present state**.



- **Step 1:**

Start with the logic schematic from which we can derive **excitation equations** for each flip-flop input.

- **Step 2:**

To obtain **next-state equations**, we insert the excitation equations into the characteristic equations. The **output equations** can be derived from the schematic:

*Flip-flop characteristic equation:*

active HIGH

$$JK : Q_{next} = J\bar{Q} + \bar{K}Q$$

$$D : Q_{next} = D$$

$$T : Q_{next} = T\bar{Q} + \bar{T}Q$$

- **Step 3:**

Once we have our output and next-state equations, we can generate the **next-state and output tables** as well as **state diagrams**.

- **Step 4:**

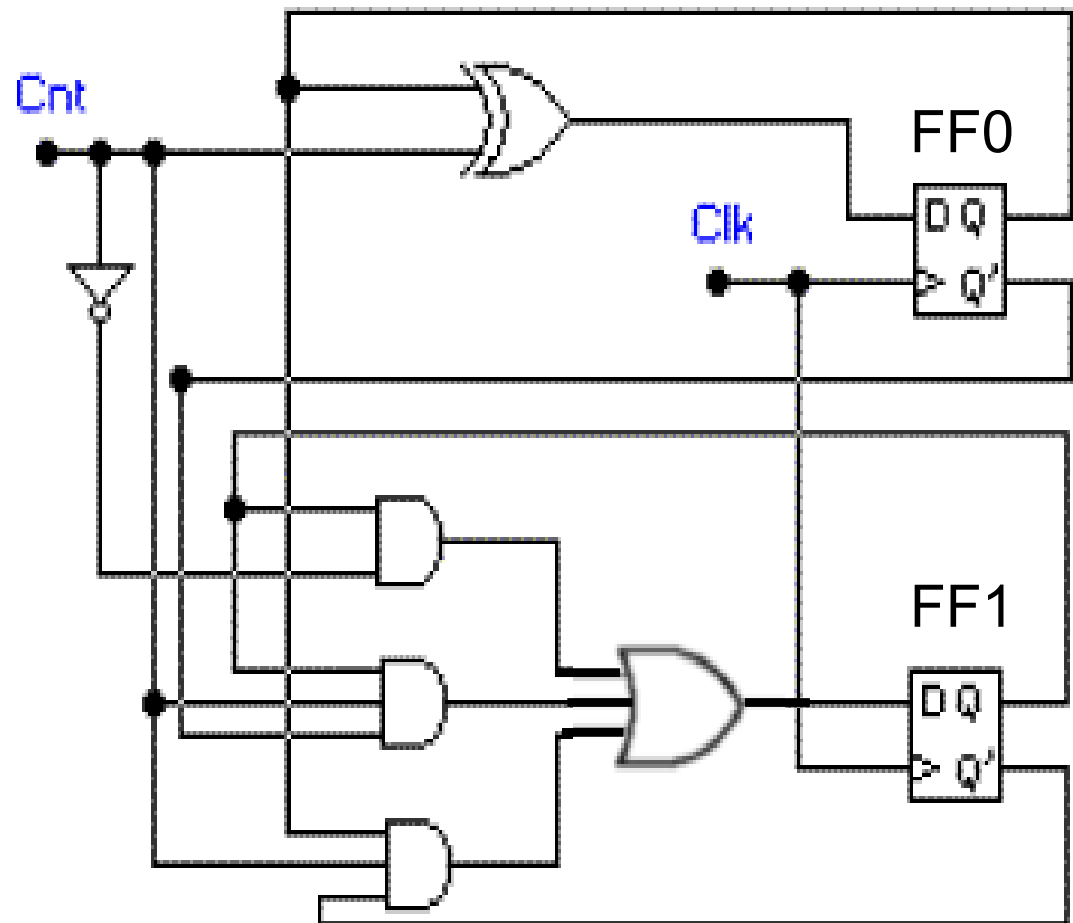
When we reach this stage, we use either the table or the state diagram to develop a **timing diagram** which can be verified through simulation.

# Analysis of Sequential Circuits:

## Modulo-4 Counter

### Example:

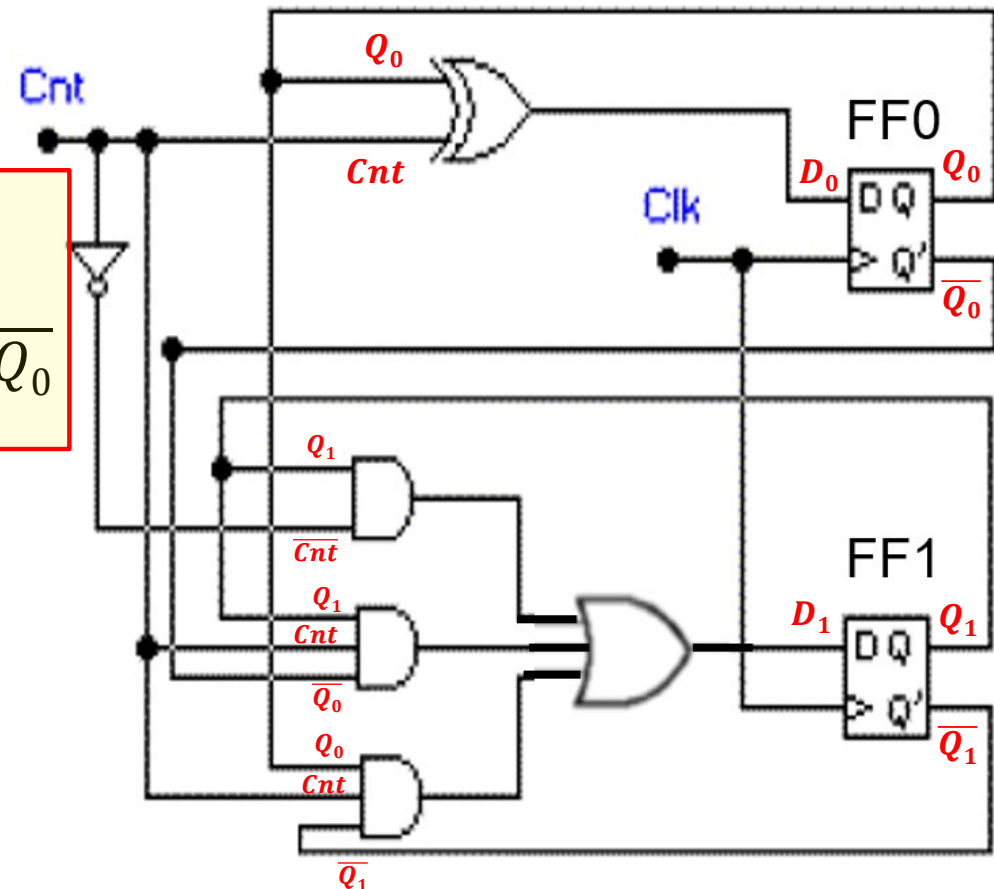
Derive the state table and state diagram for the sequential circuit below.



- **Step 1: Boolean expressions** for the inputs of each flip-flops in the schematic.

$$D_0 = Cnt \oplus Q_0 = \overline{Cnt} \cdot Q_0 + Cnt \cdot \overline{Q_0}$$

$$D_1 = \overline{Cnt} \cdot Q_1 + Cnt \cdot \overline{Q_1} \cdot Q_0 + Cnt \cdot Q_1 \cdot \overline{Q_0}$$



active HIGH

$$\text{JK : } Q_{\text{next}} = JQ + K\bar{Q}$$

$$\text{D : } Q_{\text{next}} = D$$

$$\text{T : } Q_{\text{next}} = T\bar{Q} + \bar{T}Q$$

- Step 1: **Boolean expressions** for the inputs of each flip-flops in the schematic.

$$D_0 = \text{Cnt} \oplus Q_0 = \overline{\text{Cnt}} \cdot Q_0 + \text{Cnt} \cdot \bar{Q}_0$$

$$D_1 = \overline{\text{Cnt}} \cdot Q_1 + \text{Cnt} \cdot \bar{Q}_1 \cdot Q_0 + \text{Cnt} \cdot Q_1 \cdot \bar{Q}_0$$

- Step 2:** Derive the **next-state equations** by converting these excitation equations into flip-flop characteristic equations. In the case of D flip-flops,  $Q_+ =$

$$Q_{0+} = D_0$$

$$Q_{1+} = D_1$$

active HIGH

$$\text{JK : } Q_{\text{next}} = JQ + K\bar{Q}$$

$$\text{D : } Q_{\text{next}} = D$$

$$\text{T : } Q_{\text{next}} = T\bar{Q} + \bar{T}Q$$

- Step 1: **Boolean expressions** for the inputs of each flip-flops in the schematic.

$$D_0 = \text{Cnt} \oplus Q_0 = \overline{\text{Cnt}} \cdot Q_0 + \text{Cnt} \cdot \overline{Q_0}$$

$$D_1 = \overline{\text{Cnt}} \cdot Q_1 + \text{Cnt} \cdot \overline{Q_1} \cdot Q_0 + \text{Cnt} \cdot Q_1 \cdot \overline{Q_0}$$

- Step 2:** Derive the **next-state equations** by converting these excitation equations into flip-flop characteristic equations. In the case of D flip-flops,  $Q_+ =$

$$Q_{0+} = D_0 = \text{Cnt} \oplus Q_0 = \overline{\text{Cnt}} \cdot Q_0 + \text{Cnt} \cdot \overline{Q_0}$$

$$Q_{1+} = D_1 = \overline{\text{Cnt}} \cdot Q_1 + \text{Cnt} \cdot \overline{Q_1} \cdot Q_0 + \text{Cnt} \cdot Q_1 \cdot \overline{Q_0}$$

$$D_0 = Cnt \oplus Q_0 = \overline{Cnt} \cdot Q_0 + Cnt \cdot \overline{Q_0}$$

$$D_1 = \overline{Cnt} \cdot Q_1 + Cnt \cdot \overline{Q_1} \cdot Q_0 + Cnt \cdot Q_1 \cdot \overline{Q_0}$$

- **Step 3:**  
Construct the  
**next-state table.**

| Input,<br>Cnt | Present State |       | Next State     |                |
|---------------|---------------|-------|----------------|----------------|
|               | $Q_1$         | $Q_0$ | $Q_{1+} = D_1$ | $Q_{0+} = D_0$ |
| 0             | 0             | 0     |                |                |
| 0             | 0             | 1     |                |                |
| 0             | 1             | 0     |                |                |
| 0             | 1             | 1     |                |                |
| 1             | 0             | 0     |                |                |
| 1             | 0             | 1     |                |                |
| 1             | 1             | 0     |                |                |
| 1             | 1             | 1     |                |                |

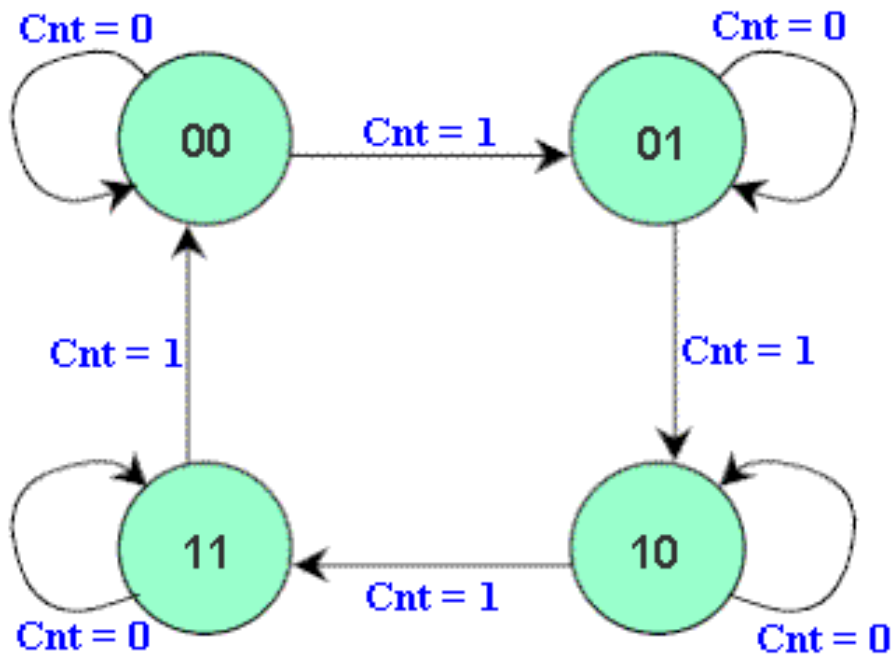
$$Q_{0+} = D_0 = Cnt \oplus Q_0 = \overline{Cnt} \cdot Q_0 + Cnt \cdot \overline{Q_0}$$

$$Q_{1+} = D_1 = \overline{Cnt} \cdot Q_1 + Cnt \cdot \overline{Q_1} \cdot Q_0 + Cnt \cdot Q_1 \cdot \overline{Q_0}$$

- Step 3:  
Construct the  
**next-state table**.

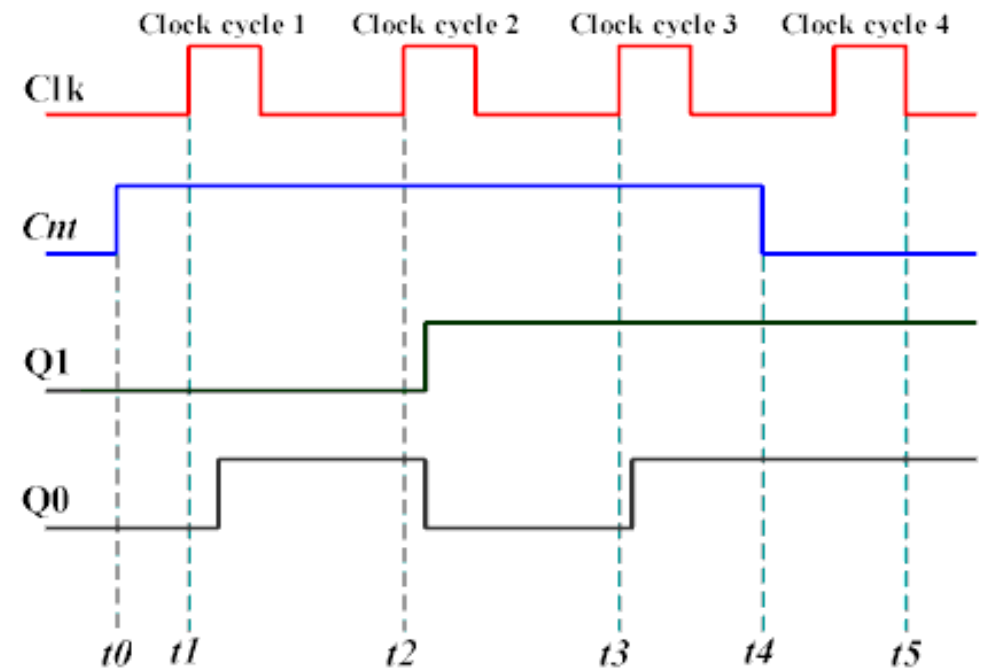
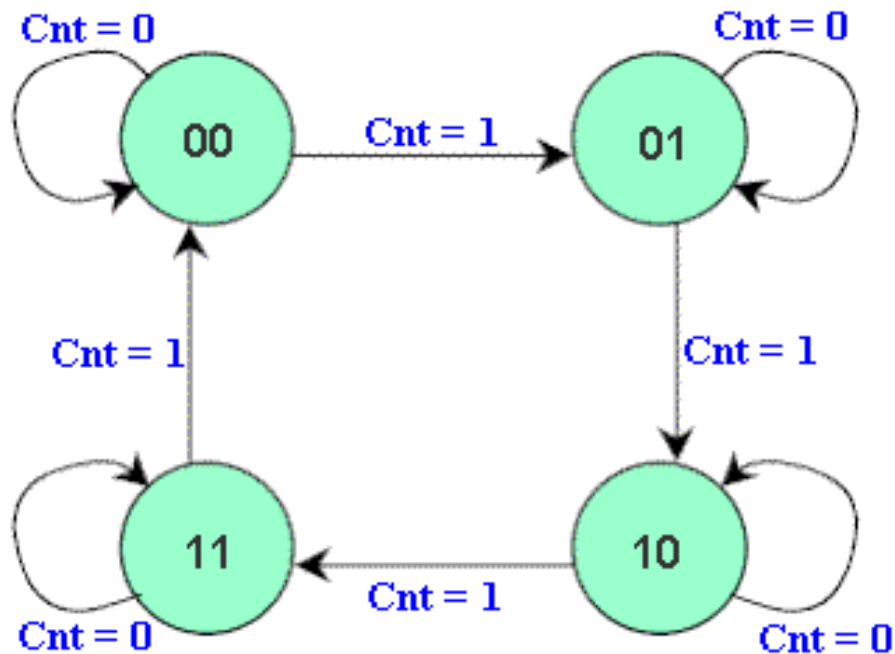
| Input,<br>Cnt | Present State |       | Next State     |                |
|---------------|---------------|-------|----------------|----------------|
|               | $Q_1$         | $Q_0$ | $Q_{1+} = D_1$ | $Q_{0+} = D_0$ |
| 0             | 0             | 0     | 0              | 0              |
| 0             | 0             | 1     | 0              | 1              |
| 0             | 1             | 0     | 1              | 0              |
| 0             | 1             | 1     | 1              | 1              |
| 1             | 0             | 0     | 0              | 1              |
| 1             | 0             | 1     | 1              | 0              |
| 1             | 1             | 0     | 1              | 1              |
| 1             | 1             | 1     | 0              | 0              |

- Step 4:
- The **state diagram** is generated directly from the next-state table.



| Input,<br>Cnt | Present State |       | Next State     |                |
|---------------|---------------|-------|----------------|----------------|
|               | $Q_1$         | $Q_0$ | $Q_{1+} = D_1$ | $Q_{0+} = D_0$ |
| 0             | 0             | 0     | 0              | 0              |
| 0             | 0             | 1     | 0              | 1              |
| 0             | 1             | 0     | 1              | 0              |
| 0             | 1             | 1     | 1              | 1              |
| 1             | 0             | 0     | 0              | 1              |
| 1             | 0             | 1     | 1              | 0              |
| 1             | 1             | 0     | 1              | 1              |
| 1             | 1             | 1     | 0              | 0              |

- **Step 4:**
- The **state diagram** is generated directly from the next-state table.
- Next get the **timing diagram**



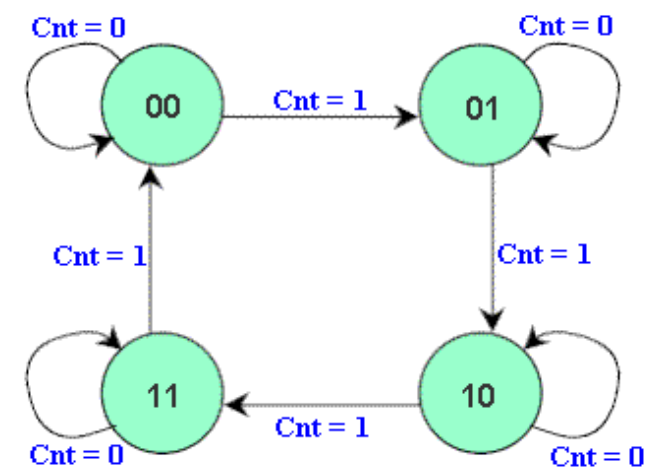
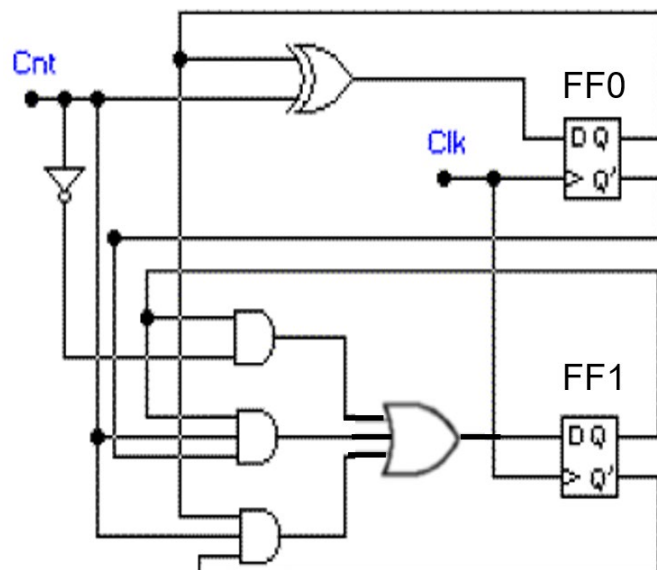
**We can conclude:**

a) From the counter circuit:

- 2-bit counter because there are 2 FFs in the design.
- Synchronous counter because the FFs have a common clock.

b) From the counter state diagram:

- MOD 4 because has 4 state (i.e.  $2^2=4$ ), not a truncated counter.
- Count Up counter when Cnt=1, and stay in the previous state when Cnt=0.

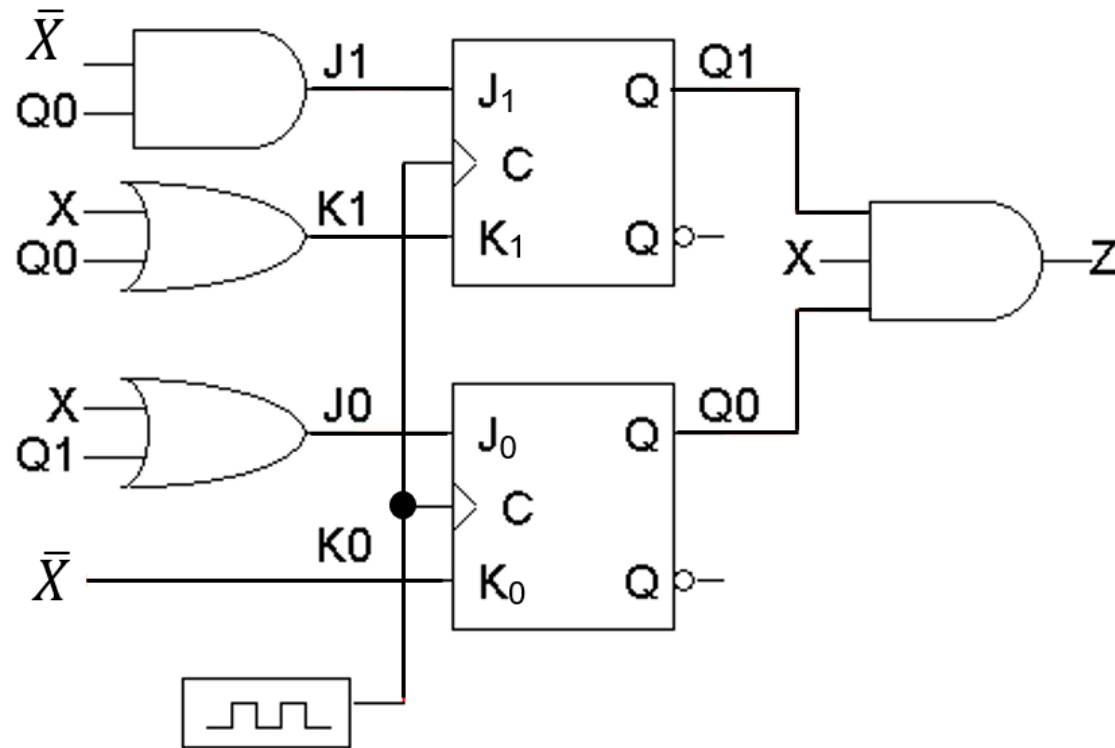


# Analysis of Sequential Circuits:

## JK Circuit Analysis

### Assume:

- Sequential circuit with two JK flip-flops.
- There is one input,  $X$ , and one output,  $Z$ .



$$\begin{aligned} J_1 &= \bar{X}Q_0 & J_0 &= X + Q_1 & Z &= Q_1Q_0X \\ K_1 &= X + Q_0 & K_0 &= \bar{X} \end{aligned}$$

- Remember that there is one input  $X$ , one output  $Z$ , and two flip-flops  $Q_1Q_0$ .
- The present state  $Q_1Q_0$  and the input will determine the next state and the output.
- From the diagram,  $Z = Q_1Q_0X$

| Input,<br>$X$ | Present State |       | Next State |          | Output,<br>$Z$ |
|---------------|---------------|-------|------------|----------|----------------|
|               | $Q_1$         | $Q_0$ | $Q_{1+}$   | $Q_{0+}$ |                |
| 0             | 0             | 0     |            |          |                |
| 0             | 0             | 1     |            |          |                |
| 0             | 1             | 0     |            |          |                |
| 0             | 1             | 1     |            |          |                |
| 1             | 0             | 0     |            |          |                |
| 1             | 0             | 1     |            |          |                |
| 1             | 1             | 0     |            |          |                |
| 1             | 1             | 1     |            |          |                |

\*\*Fill in the Z

- Remember that there is one input  $X$ , one output  $Z$ , and two flip-flops  $Q_1Q_0$ .
- The present state  $Q_1Q_0$  and the input will determine the next state and the output.
- From the diagram,  $Z = Q_1Q_0X$

| Input,<br>$X$ | Present State |       | Next State |          | Output,<br>$Z$ |
|---------------|---------------|-------|------------|----------|----------------|
|               | $Q_1$         | $Q_0$ | $Q_{1+}$   | $Q_{0+}$ |                |
| 0             | 0             | 0     |            |          | 0              |
| 0             | 0             | 1     |            |          | 0              |
| 0             | 1             | 0     |            |          | 0              |
| 0             | 1             | 1     |            |          | 0              |
| 1             | 0             | 0     |            |          | 0              |
| 1             | 0             | 1     |            |          | 0              |
| 1             | 1             | 0     |            |          | 0              |
| 1             | 1             | 1     |            |          | 1              |

\*\*Fill in the Z

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

active HIGH SR :  $Q_{\text{next}} = S + \bar{R}Q, SR = 0$

$$\text{JK : } Q_{\text{next}} = J\bar{Q} + \bar{K}Q$$

$$\text{D : } Q_{\text{next}} = D$$

$$\text{T : } Q_{\text{next}} = T\bar{Q} + \bar{T}Q$$

$$Q_{0+} = J_0\bar{Q}_0 + \bar{K}_0Q_0 = (X + Q_1)\bar{Q}_0 + \bar{X}Q_0 = X\bar{Q}_0 + Q_1\bar{Q}_0 + XQ_0$$

$$Q_{1+} = J_1\bar{Q}_1 + \bar{K}_1Q_1 = (\bar{X}Q_0)\bar{Q}_1 + (\bar{X} + \bar{Q}_0)Q_1 = \bar{X}Q_0\bar{Q}_1 + (\bar{X} + \bar{Q}_0)Q_1 = \bar{X}Q_0\bar{Q}_1 + \bar{X}\bar{Q}_0Q_1 + \bar{Q}_0Q_1$$

$$= \bar{X}(Q_0\bar{Q}_1 + \bar{Q}_0Q_1) + \bar{Q}_0Q_1$$

| Input,<br>X | Present State  |                | Next State      |                 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |              |
| 0           | 0              | 0              |                 |                 | 0            |
| 0           | 0              | 1              |                 |                 | 0            |
| 0           | 1              | 0              |                 |                 | 0            |
| 0           | 1              | 1              |                 |                 | 0            |
| 1           | 0              | 0              |                 |                 | 0            |
| 1           | 0              | 1              |                 |                 | 0            |
| 1           | 1              | 0              |                 |                 | 0            |
| 1           | 1              | 1              |                 |                 | 1            |

**\*\*Fill in the next state column**

active HIGH SR :  $Q_{next} = S + \bar{R}Q, SR = 0$

$$JK : Q_{next} = J\bar{Q} + \bar{K}Q$$

$$D : Q_{next} = D$$

$$T : Q_{next} = T\bar{Q} + \bar{T}Q$$

$$Q_{0+} = X\bar{Q}_0 + Q_1\bar{Q}_0 + XQ_0$$

$$Q_{1+} = \bar{X}(Q_0\bar{Q}_1 + \bar{Q}_0Q_1)$$

| Input,<br>X | Present State |       | Next State |          | Output,<br>Z |
|-------------|---------------|-------|------------|----------|--------------|
|             | $Q_1$         | $Q_0$ | $Q_{1+}$   | $Q_{0+}$ |              |
| 0           | 0             | 0     | 0          | 0        | 0            |
| 0           | 0             | 1     | 1          | 0        | 0            |
| 0           | 1             | 0     | 1          | 1        | 0            |
| 0           | 1             | 1     | 0          | 0        | 0            |
| 1           | 0             | 0     | 0          | 1        | 0            |
| 1           | 0             | 1     | 0          | 1        | 0            |
| 1           | 1             | 0     | 0          | 1        | 0            |
| 1           | 1             | 1     | 0          | 1        | 1            |

**\*\*Fill in the next state column**

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              |                 |                 |    |    |    |    | 0            |
| 0           | 0              | 1              |                 |                 |    |    |    |    | 0            |
| 0           | 1              | 0              |                 |                 |    |    |    |    | 0            |
| 0           | 1              | 1              |                 |                 |    |    |    |    | 0            |
| 1           | 0              | 0              |                 |                 |    |    |    |    | 0            |
| 1           | 0              | 1              |                 |                 |    |    |    |    | 0            |
| 1           | 1              | 0              |                 |                 |    |    |    |    | 0            |
| 1           | 1              | 1              |                 |                 |    |    |    |    | 1            |

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              |                 |                 | 0  |    |    |    | 0            |
| 0           | 0              | 1              |                 |                 | 1  |    |    |    | 0            |
| 0           | 1              | 0              |                 |                 | 0  |    |    |    | 0            |
| 0           | 1              | 1              |                 |                 | 1  |    |    |    | 0            |
| 1           | 0              | 0              |                 |                 | 0  |    |    |    | 0            |
| 1           | 0              | 1              |                 |                 | 0  |    |    |    | 0            |
| 1           | 1              | 0              |                 |                 | 0  |    |    |    | 0            |
| 1           | 1              | 1              |                 |                 | 0  |    |    |    | 1            |

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              |                 |                 | 0  | 0  |    |    | 0            |
| 0           | 0              | 1              |                 |                 | 1  | 1  |    |    | 0            |
| 0           | 1              | 0              |                 |                 | 0  | 0  |    |    | 0            |
| 0           | 1              | 1              |                 |                 | 1  | 1  |    |    | 0            |
| 1           | 0              | 0              |                 |                 | 0  | 1  |    |    | 0            |
| 1           | 0              | 1              |                 |                 | 0  | 1  |    |    | 0            |
| 1           | 1              | 0              |                 |                 | 0  | 1  |    |    | 0            |
| 1           | 1              | 1              |                 |                 | 0  | 1  |    |    | 1            |

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              |                 |                 | 0  | 0  | 0  |    | 0            |
| 0           | 0              | 1              |                 |                 | 1  | 1  | 0  |    | 0            |
| 0           | 1              | 0              |                 |                 | 0  | 0  | 1  |    | 0            |
| 0           | 1              | 1              |                 |                 | 1  | 1  | 1  |    | 0            |
| 1           | 0              | 0              |                 |                 | 0  | 1  | 1  |    | 0            |
| 1           | 0              | 1              |                 |                 | 0  | 1  | 1  |    | 0            |
| 1           | 1              | 0              |                 |                 | 0  | 1  | 1  |    | 0            |
| 1           | 1              | 1              |                 |                 | 0  | 1  | 1  |    | 1            |

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              |                 |                 | 0  | 0  | 0  | 1  | 0            |
| 0           | 0              | 1              |                 |                 | 1  | 1  | 0  | 1  | 0            |
| 0           | 1              | 0              |                 |                 | 0  | 0  | 1  | 1  | 0            |
| 0           | 1              | 1              |                 |                 | 1  | 1  | 1  | 1  | 0            |
| 1           | 0              | 0              |                 |                 | 0  | 1  | 1  | 0  | 0            |
| 1           | 0              | 1              |                 |                 | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 0              |                 |                 | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 1              |                 |                 | 0  | 1  | 1  | 0  | 1            |

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              |                 |                 | 0  | 0  | 0  | 1  | 0            |
| 0           | 0              | 1              |                 |                 | 1  | 1  | 0  | 1  | 0            |
| 0           | 1              | 0              |                 |                 | 0  | 0  | 1  | 1  | 0            |
| 0           | 1              | 1              |                 |                 | 1  | 1  | 1  | 1  | 0            |
| 1           | 0              | 0              |                 |                 | 0  | 1  | 1  | 0  | 0            |
| 1           | 0              | 1              |                 |                 | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 0              |                 |                 | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 1              |                 |                 | 0  | 1  | 1  | 0  | 1            |

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              | 0               | 0               | 0  | 0  | 0  | 1  | 0            |
| 0           | 0              | 1              | 1               | 0               | 1  | 1  | 0  | 1  | 0            |
| 0           | 1              | 0              | 1               | 1               | 0  | 0  | 1  | 1  | 0            |
| 0           | 1              | 1              | 0               | 0               | 1  | 1  | 1  | 1  | 0            |
| 1           | 0              | 0              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 0              | 1              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 0              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 1              | 0               | 1               | 0  | 1  | 1  | 0  | 1            |

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              | 0               | 0               | 0  | 0  | 0  | 1  | 0            |
| 0           | 0              | 1              | 1               | 0               | 1  | 1  | 0  | 1  | 0            |
| 0           | 1              | 0              | 1               | 1               | 0  | 0  | 1  | 1  | 0            |
| 0           | 1              | 1              | 0               | 0               | 1  | 1  | 1  | 1  | 0            |
| 1           | 0              | 0              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 0              | 1              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 0              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 1              | 0               | 1               | 0  | 1  | 1  | 0  | 1            |

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              | 0               | 0               | 0  | 0  | 0  | 1  | 0            |
| 0           | 0              | 1              | 1               | 0               | 1  | 1  | 0  | 1  | 0            |
| 0           | 1              | 0              | 1               | 1               | 0  | 0  | 1  | 1  | 0            |
| 0           | 1              | 1              | 0               | 0               | 1  | 1  | 1  | 1  | 0            |
| 1           | 0              | 0              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 0              | 1              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 0              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 1              | 0               | 1               | 0  | 1  | 1  | 0  | 1            |

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              | 0               | 0               | 0  | 0  | 0  | 1  | 0            |
| 0           | 0              | 1              | 1               | 0               | 1  | 1  | 0  | 1  | 0            |
| 0           | 1              | 0              | 1               | 1               | 0  | 0  | 1  | 1  | 0            |
| 0           | 1              | 1              | 0               | 0               | 1  | 1  | 1  | 1  | 0            |
| 1           | 0              | 0              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 0              | 1              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 0              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 1              | 0               | 1               | 0  | 1  | 1  | 0  | 1            |

## Method 2 –

- Fill up JK input values based on input equations
- Fill up Next State columns by observing JK input values

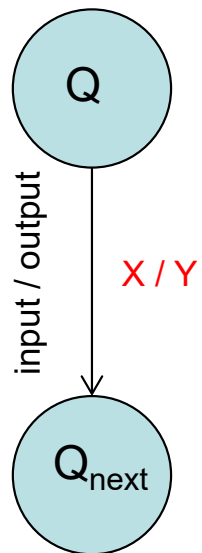
$$J_1 = \bar{X}Q_0 \quad J_0 = X + Q_1 \quad Z = Q_1Q_0X$$

$$K_1 = X + Q_0 \quad K_0 = \bar{X}$$

| Input,<br>X | Present State  |                | Next State      |                 | J1 | K1 | J0 | K0 | Output,<br>Z |
|-------------|----------------|----------------|-----------------|-----------------|----|----|----|----|--------------|
|             | Q <sub>1</sub> | Q <sub>0</sub> | Q <sub>1+</sub> | Q <sub>0+</sub> |    |    |    |    |              |
| 0           | 0              | 0              | 0               | 0               | 0  | 0  | 0  | 1  | 0            |
| 0           | 0              | 1              | 1               | 0               | 1  | 1  | 0  | 1  | 0            |
| 0           | 1              | 0              | 1               | 1               | 0  | 0  | 1  | 1  | 0            |
| 0           | 1              | 1              | 0               | 0               | 1  | 1  | 1  | 1  | 0            |
| 1           | 0              | 0              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 0              | 1              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 0              | 0               | 1               | 0  | 1  | 1  | 0  | 0            |
| 1           | 1              | 1              | 0               | 1               | 0  | 1  | 1  | 0  | 1            |

**Exercise 8b.20:** Draw the state diagram for the example in previous slide.

Present state



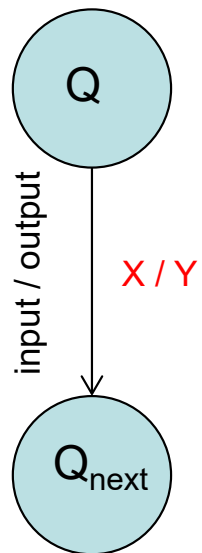
Next state

| Input,<br>X | Present State |       | Next State |          | Output,<br>Z |
|-------------|---------------|-------|------------|----------|--------------|
|             | $Q_1$         | $Q_0$ | $Q_{1+}$   | $Q_{0+}$ |              |
| 0           | 0             | 0     | 0          | 0        | 0            |
| 0           | 0             | 1     | 1          | 0        | 0            |
| 0           | 1             | 0     | 1          | 1        | 0            |
| 0           | 1             | 1     | 0          | 0        | 0            |
| 1           | 0             | 0     | 0          | 1        | 0            |
| 1           | 0             | 1     | 0          | 1        | 0            |
| 1           | 1             | 0     | 0          | 1        | 0            |
| 1           | 1             | 1     | 0          | 1        | 1            |

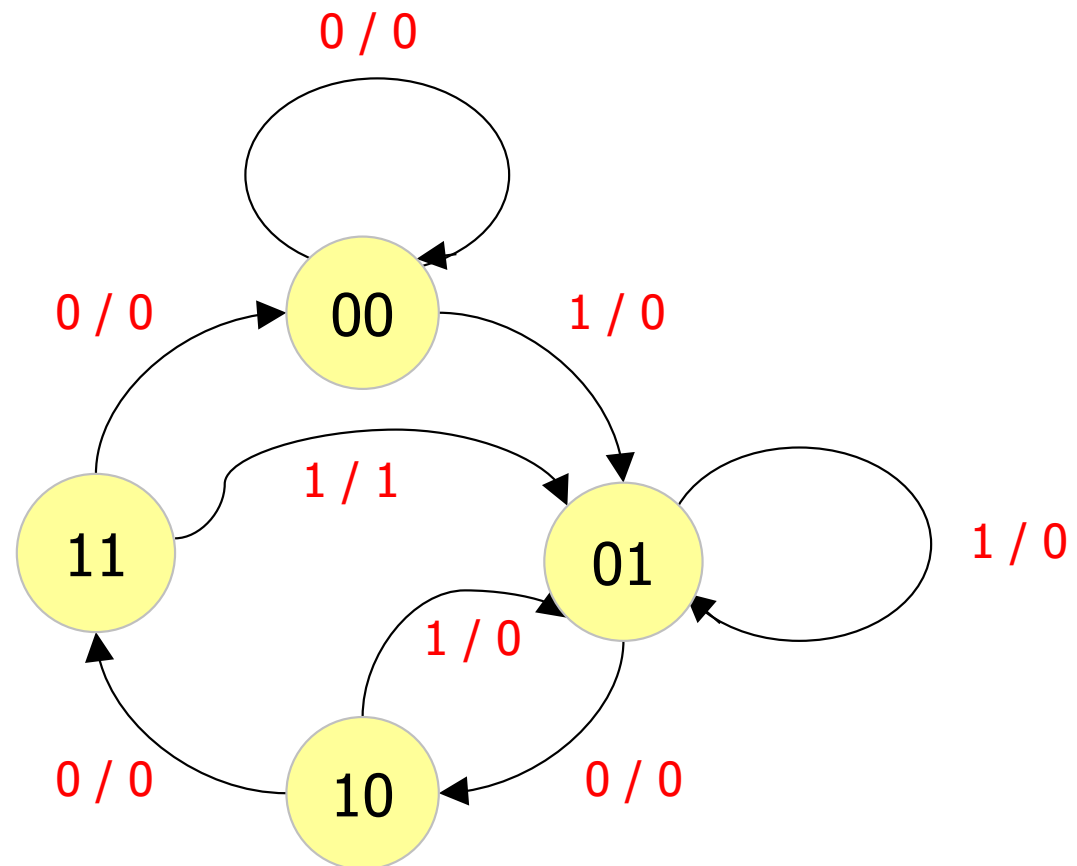
## Solution :

Draw the state diagram for the example in previous slide.

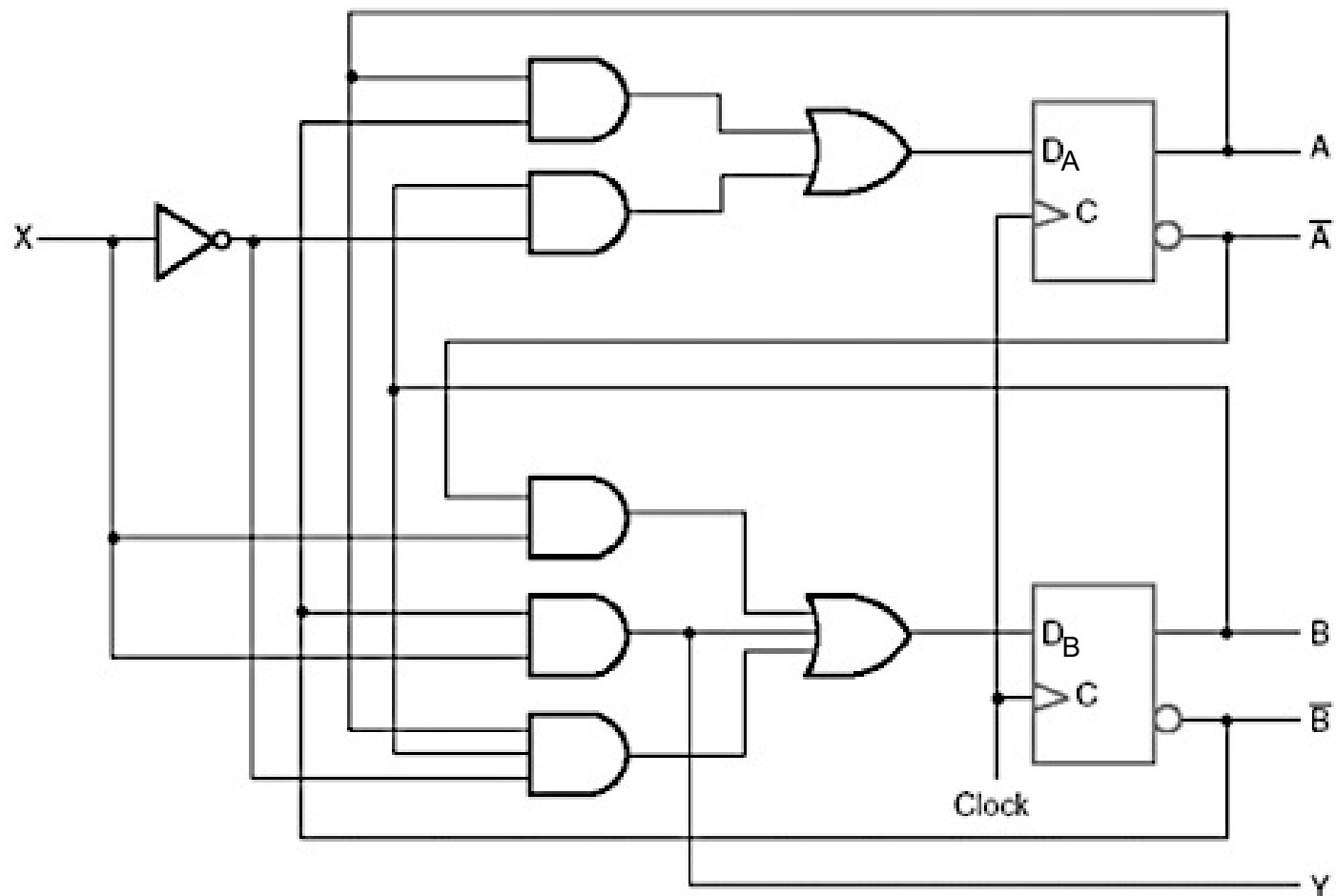
Present state



Next state



**Exercise 8b.21:** Analysis for the following sequential circuit.



## Solution:

**Step 1:** The equation of each FF and output, Y

$$D_A = \bar{X}B + A\bar{B}$$

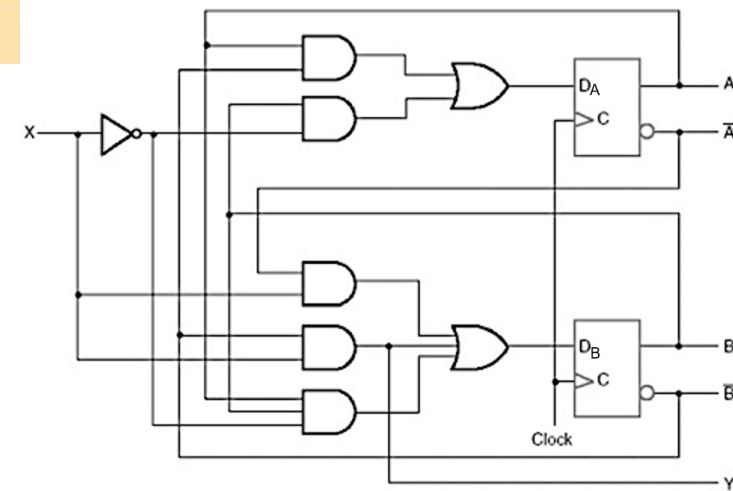
$$D_B = \bar{A}X + \bar{B}X + AB\bar{X}$$

$$Y = \bar{B}X$$

**Step 2:** Derive the next state equations by converting these excitation equation into FF characteristic equations.

$$D_{A+} = D_A$$

$$D_{B+} = D_B$$



active HIGH SR :  $Q_{next} = S + \bar{R}Q$ ,  $SR = 0$   
JK :  $Q_{next} = J\bar{Q} + \bar{K}Q$   
**D :  $Q_{next} = D$**   
T :  $Q_{next} = T\bar{Q} + \bar{T}Q$

## Solution:

**Step 1:** The equation of each FF and output, Y

$$D_A = \bar{X}B + A\bar{B}$$

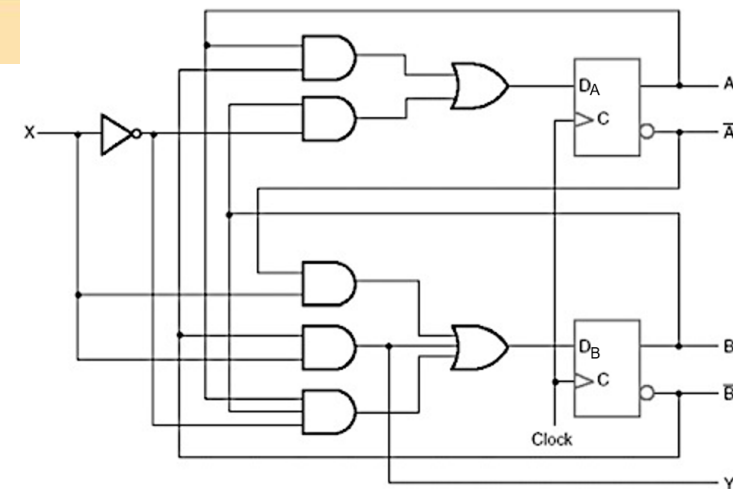
$$D_B = \bar{A}X + \bar{B}X + AB\bar{X}$$

$$Y = \bar{B}X$$

**Step 2:** Derive the next state equations by converting these excitation equation into FF characteristic equations.

$$D_{A+} = D_A = \bar{X}B + A\bar{B}$$

$$D_{B+} = D_B = \bar{A}X + \bar{B}X + AB\bar{X}$$



active HIGH SR :  $Q_{next} = S + \bar{R}Q$ ,  $SR = 0$   
JK :  $Q_{next} = J\bar{Q} + \bar{K}Q$   
**D :  $Q_{next} = D$**   
T :  $Q_{next} = T\bar{Q} + \bar{T}Q$

**Step 3:** Produce a Next State Table.

$$D_{A+} = D_A = \overline{X}B + A\overline{B}$$

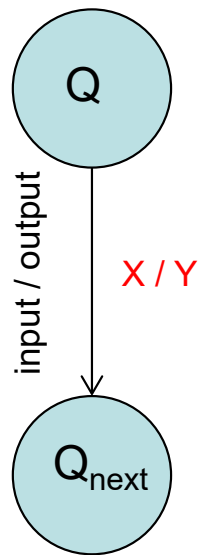
$$D_{B+} = D_B = \overline{A}X + \overline{B}X + AB\overline{X}$$

$$Y = \overline{B}X$$

| Input,<br>X | Present State |   | Next State     |                | Output,<br>Y |
|-------------|---------------|---|----------------|----------------|--------------|
|             | A             | B | A <sub>+</sub> | B <sub>+</sub> |              |
| 0           | 0             | 0 | 0              | 0              | 0            |
| 0           | 0             | 1 | 1              | 0              | 0            |
| 0           | 1             | 0 | 1              | 0              | 0            |
| 0           | 1             | 1 | 1              | 1              | 0            |
| 1           | 0             | 0 | 0              | 1              | 1            |
| 1           | 0             | 1 | 0              | 1              | 0            |
| 1           | 1             | 0 | 1              | 1              | 1            |
| 1           | 1             | 1 | 0              | 0              | 0            |

**Exercise 8b.22:** Draw the state diagram for the example in previous slide.

Present state



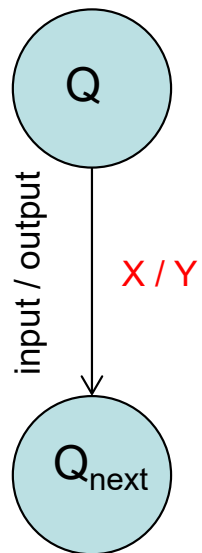
Next state

| Input,<br>X | Present State |   | Next State     |                | Output,<br>Y |
|-------------|---------------|---|----------------|----------------|--------------|
|             | A             | B | A <sub>+</sub> | B <sub>+</sub> |              |
| 0           | 0             | 0 | 0              | 0              | 0            |
| 0           | 0             | 1 | 1              | 0              | 0            |
| 0           | 1             | 0 | 1              | 0              | 0            |
| 0           | 1             | 1 | 1              | 1              | 0            |
| 1           | 0             | 0 | 0              | 1              | 1            |
| 1           | 0             | 1 | 0              | 1              | 0            |
| 1           | 1             | 0 | 1              | 1              | 1            |
| 1           | 1             | 1 | 0              | 0              | 0            |

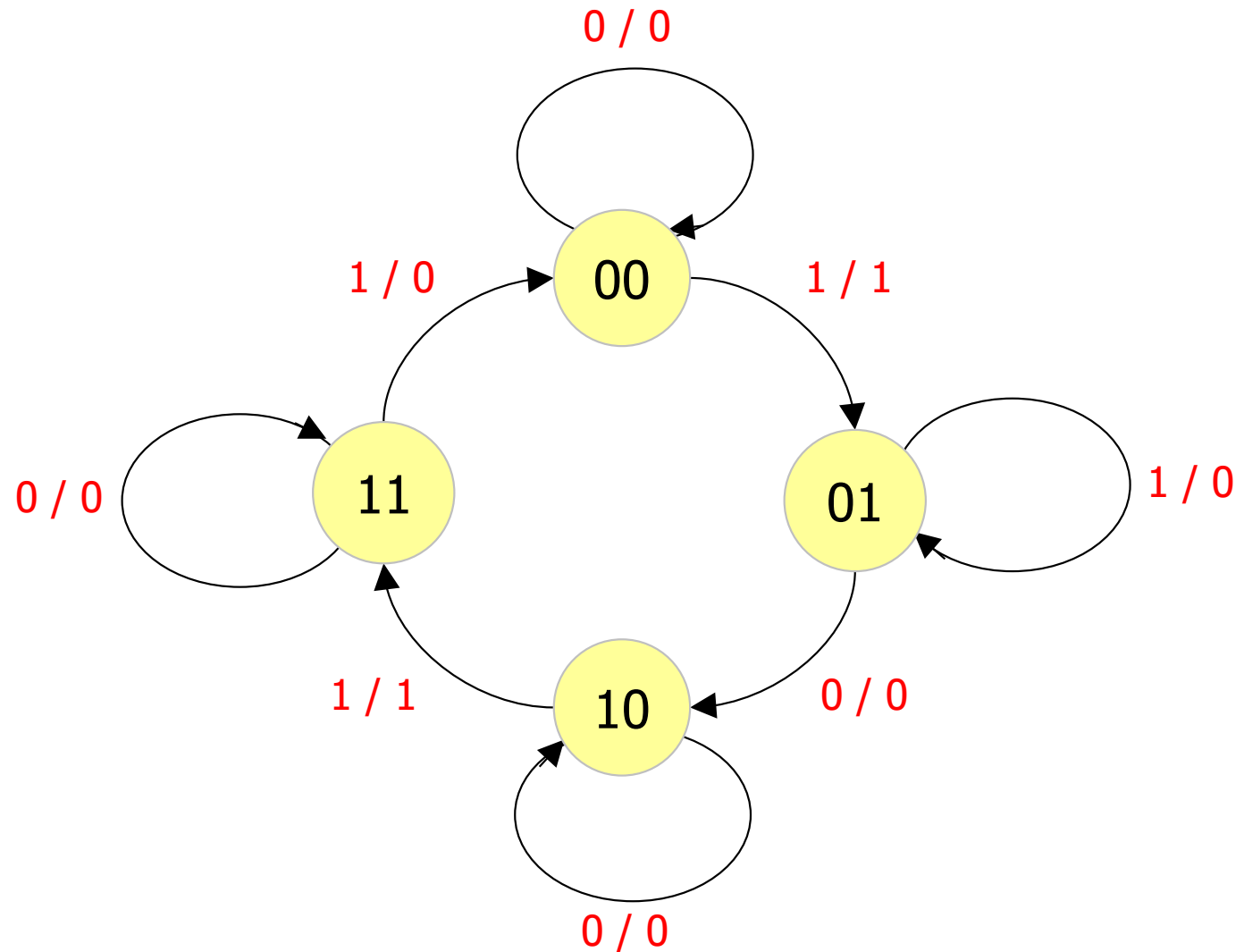
## Solution :

Draw the state diagram for the example in previous slide.

Present state



Next state

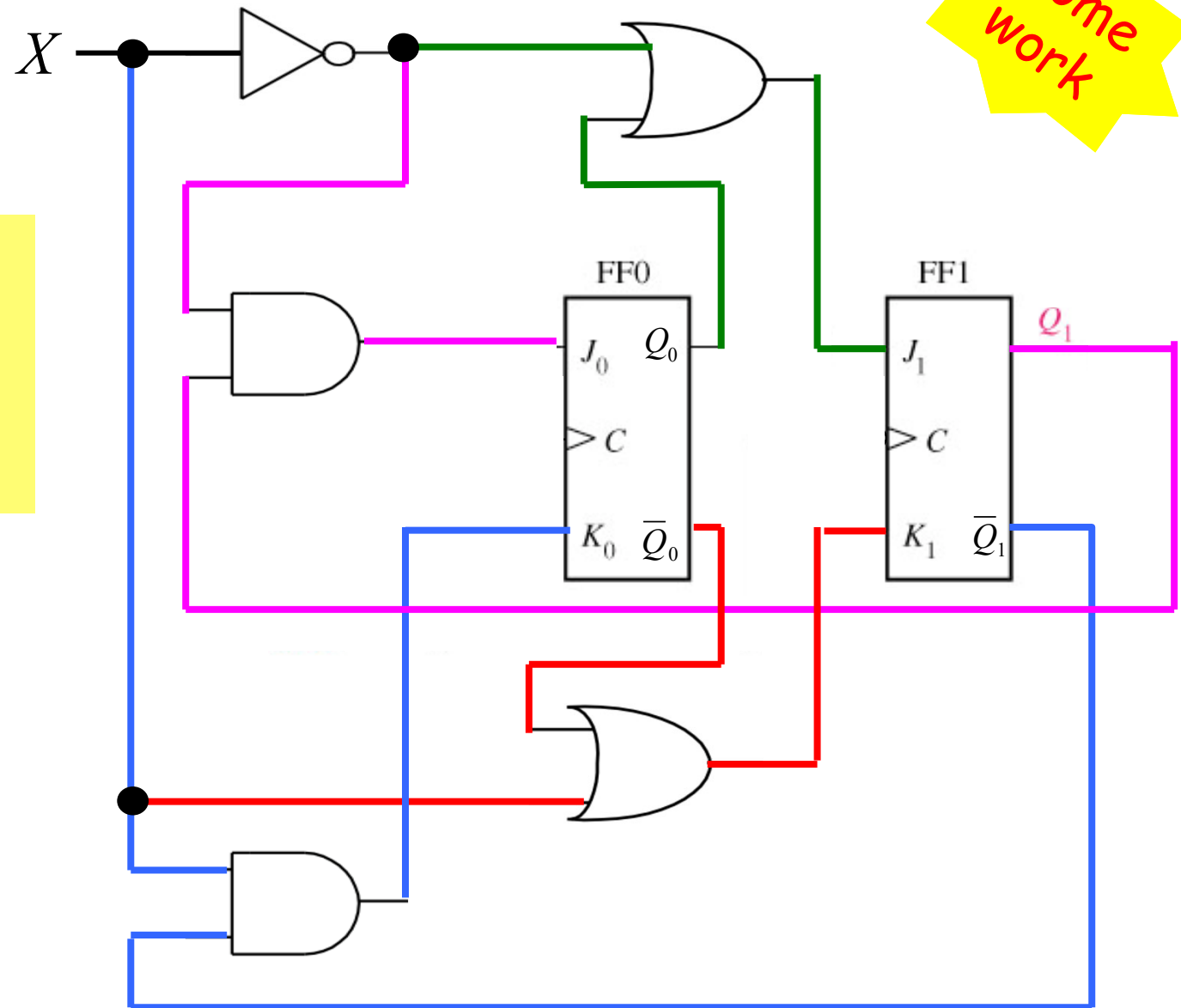


# End of Module 8

Home  
work

**Exercise 8b.23:**

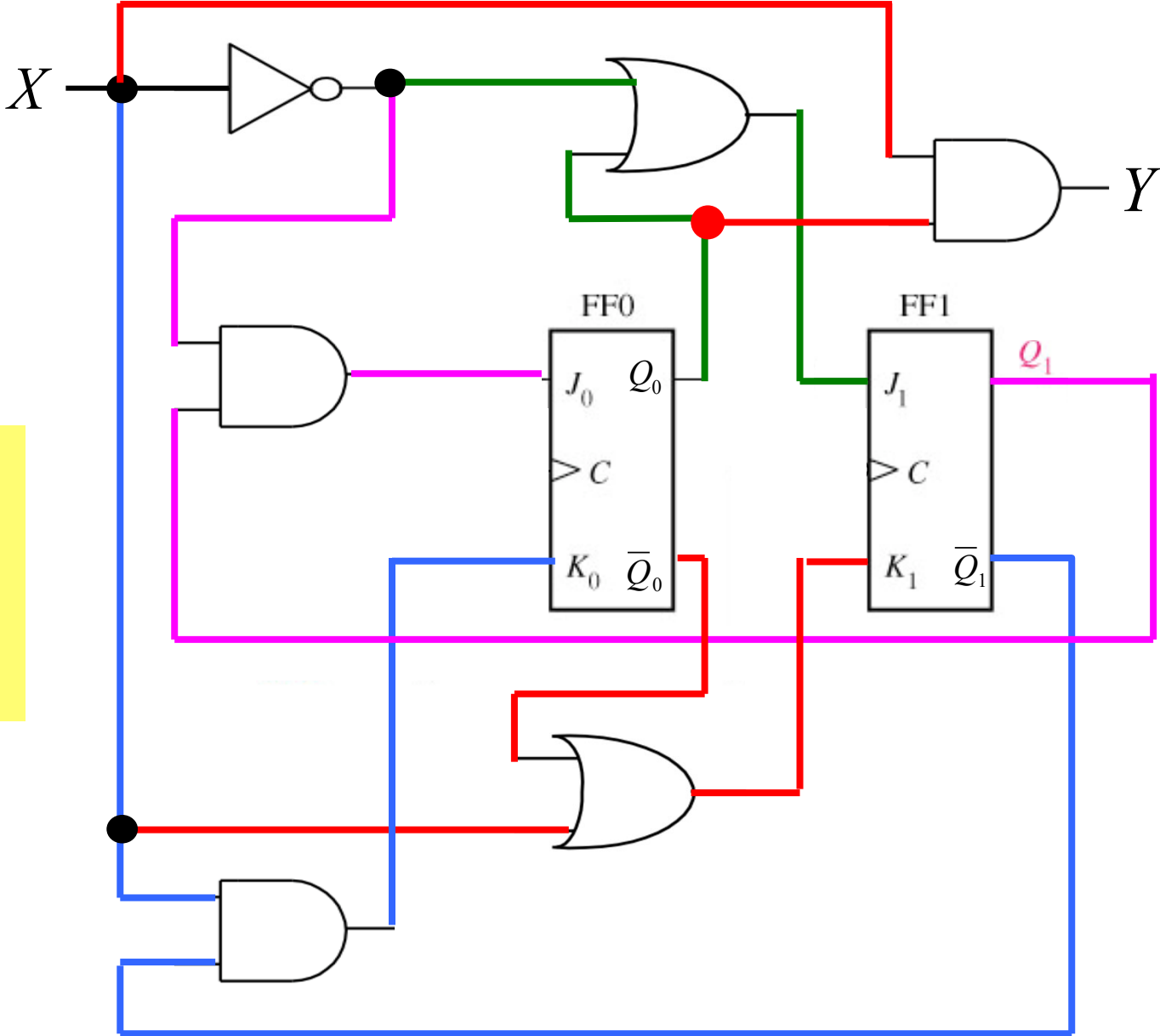
Derive the state table and state diagram for the sequential circuit below.





## Exercise 8b.24:

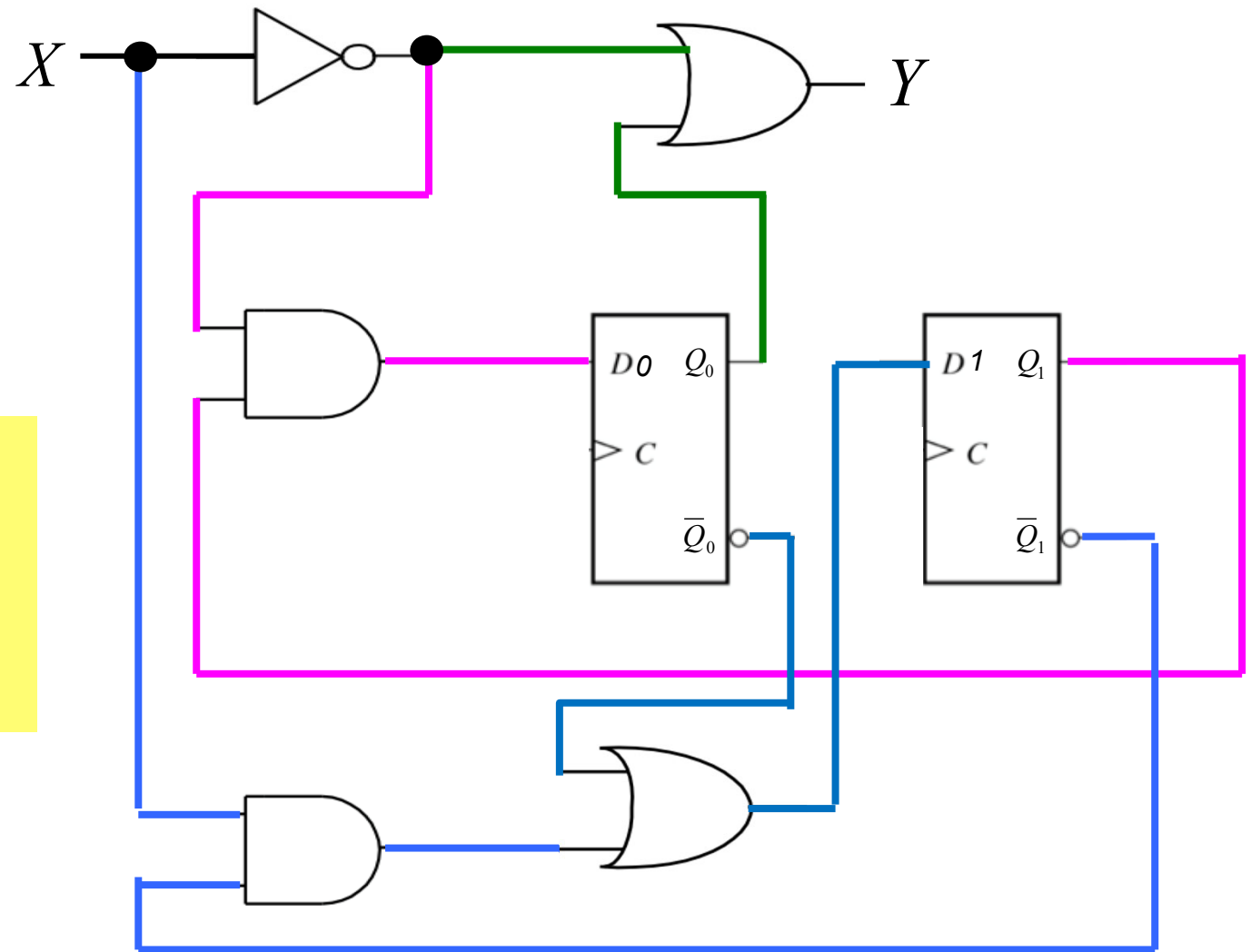
Derive the state table and state diagram for the sequential circuit below.





### Exercise 8b.25:

Derive the state table and state diagram for the sequential circuit below.





### Exercise 8b.27:

Derive the state table and state diagram for the sequential circuit below.

